



THE OHIO STATE UNIVERSITY

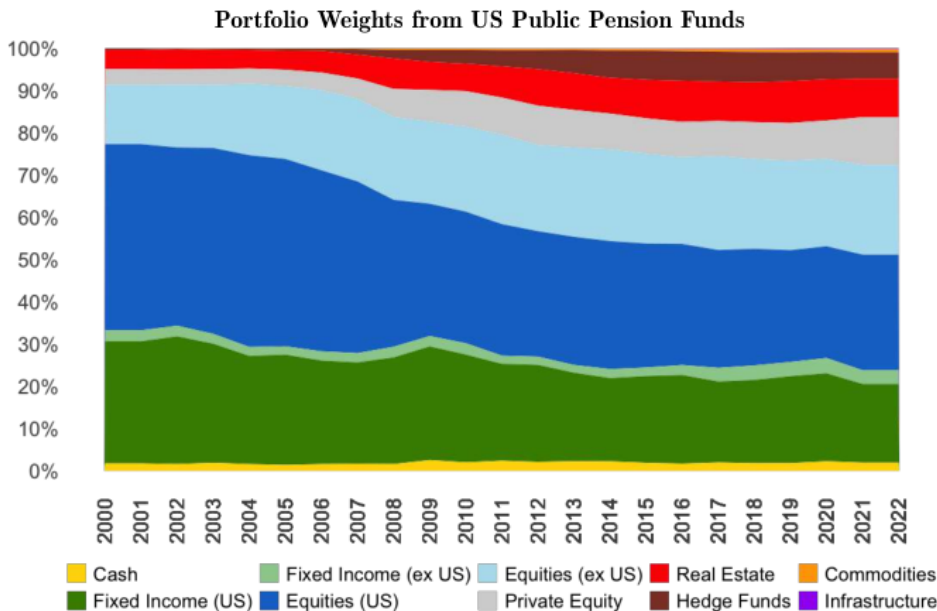
FISHER COLLEGE OF BUSINESS

Variation in Asset Prices & Expected Returns

(BUSFIN 8200 - Asset Pricing Foundations)

Andrei S. Gonçalves

Demand for Financial Assets: The Case of Pension Funds



Source: Coutts, Gonçalves, Liu, Loudis (2024)

- Theories developed in this course focus on the demand for financial assets
 - * But the supply of financial assets also matters!
- Evidence covered in this course focuses on Equities and Bonds (most of Fixed Income)
 - * But other asset classes also matter!

This Module (Variation in Asset Prices & $\mathbb{E}[R]$)

(1) Variation in Equity Prices

[1.1] The Present Value Identity

$$p_{j,t} = cf_{j,t} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta cf_{j,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{j,t+h}]$$

[1.2] Variation in Aggregate Equity Prices

[1.3] Variation in Firm-Level Equity Prices

[1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

[2.1] Equity Premium Predictors

[2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

[3.1] Equity Anomaly Signals

[3.2] Important Issues and Developments

(4) Bond Prices and $\mathbb{E}[R]$

[4.1] Treasury Bonds

[4.2] Corporate Bonds

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

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(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

(1) Variation in Equity Prices

[1.1] The Present Value Identity

- $\uparrow CF_t \implies \uparrow P_t$
- $\uparrow \mathbb{E}_t[\text{Growth}] \implies \uparrow P_t$
- $\uparrow \mathbb{E}_t[R] \implies \downarrow P_t$

[1.2] Variation in Aggregate Equity Prices

- Secular movements in aggregate equity prices are driven by CF_t
- Deviations of aggregate P_t from CF_t are mostly driven by $\mathbb{E}_t[R]$

[1.3] Variation in Firm-Level Equity Prices

- Focus on market equity (ME_t) and book equity (BE_t) to avoid issues with $CF_t \leq 0$
- Deviations of ME_t from BE_t across firms are mostly driven by $\mathbb{E}_t[prof]$
- At the firm-level, shocks to returns are mostly driven by $\mathbb{E}_t[prof]$
- At the aggregate level, shocks to returns are mostly driven by $\mathbb{E}_t[R]$
- Cash flow shocks diversify: $\mathbb{E}_t[R]$ is more relevant for aggregate returns

[1.4] Important Issues and Developments

- Sample period, empirical specification, and measurement of dividends matter
- How we normalize prices matter (empirically and conceptually)

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[4.1] Treasury Bonds

[4.2] Corporate Bonds

The Present Value Identity

- From the definition of a gross return (and suppressing the j index):

$$R_{t+1} = \frac{P_{t+1} + CF_{t+1}}{P_t}$$

$$\Downarrow$$

$$\mathbb{E}_t[R] = \frac{\mathbb{E}_t[P_{t+1} + CF_{t+1}]}{P_t}$$

- Isolating P_t and recursively substituting P_{t+h} :

$$P_t = \mathbb{E}_t \left[\frac{P_{t+1} + CF_{t+1}}{\mathbb{E}_t[R]} \right]$$

$$= \mathbb{E}_t \left[\frac{CF_{t+1}}{\mathbb{E}_t[R]} \right] + \mathbb{E}_t \left[\frac{P_{t+1} + CF_{t+2}}{\mathbb{E}_t[R] \cdot \mathbb{E}_{t+1}[R]} \right]$$

$$= \mathbb{E}_t \left[\frac{CF_{t+1}}{\mathbb{E}_t[R]} \right] + \mathbb{E}_t \left[\frac{CF_{t+2}}{\mathbb{E}_t[R] \cdot \mathbb{E}_{t+1}[R]} \right] + \mathbb{E}_t \left[\frac{P_{t+3} + CF_{t+3}}{\mathbb{E}_t[R] \cdot \mathbb{E}_{t+1}[R] \cdot \mathbb{E}_{t+2}[R]} \right]$$

$$\vdots \quad \text{keep substituting } P_{t+h}$$

$$= \mathbb{E}_t \left[\frac{CF_{t+1}}{\mathbb{E}_t[R]} \right] + \mathbb{E}_t \left[\frac{CF_{t+2}}{\mathbb{E}_t[R] \cdot \mathbb{E}_{t+1}[R]} \right] + \dots + \lim_{h \rightarrow \infty} \mathbb{E}_t \left[\frac{P_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

The No-Bubbles Condition

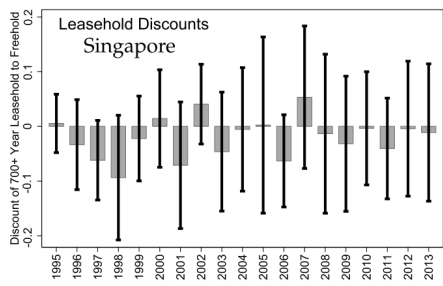
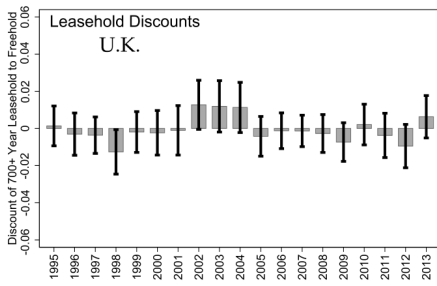
$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{\textcolor{green}{CF}_{t+h}}{\prod_{\tau=0}^{h-1} \textcolor{red}{E}_{t+\tau}[\textcolor{red}{R}]}} \right] \quad \text{requires} \quad \lim_{h \rightarrow \infty} \mathbb{E}_t \left[\frac{P_{t+h}}{\prod_{\tau=0}^{h-1} \textcolor{red}{E}_{t+\tau}[\textcolor{red}{R}]}} \right] = 0$$

- Rational Bubbles:
 - Refer to bubbles under objective $\mathbb{E}[\cdot]$ (would exist in reality)
 - Have restrictive existence conditions ▶ (Details)
 - Can be tested using price data
- Irrational Bubbles:
 - Refer to bubbles under subjective $\mathbb{E}[\cdot]$ (would only exist in investors' mind)
 - Have flexible existence conditions
 - Can only be tested with data on investors' beliefs
- So, empirical tests focus on rational bubbles
- Traditional empirical tests have mixed results:
 - Test whether P_t and $\textcolor{green}{CF}_t$ are cointegrated
 - If P_t deviates from $\textcolor{green}{CF}_t$ in an explosive fashion, then we have a rational bubble
 - So, these tests rely on econometric models of stationarity
 - But it is hard to differentiate a persistent time-series from a non-stationary one
 - So, traditional empirical tests have mixed results
(e.g., Diba, Grossman (1988), Evans (1991), Phillips, Shi, Yu (2015), Engsted, Hviid, Pedersen (2016), Caravella, Psaradakis, Sola (2023))

The No-Bubbles Condition

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right] \quad \text{requires} \quad \lim_{h \rightarrow \infty} \mathbb{E}_t \left[\frac{P_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right] = 0$$

- Rational Bubbles:
 - Refer to bubbles under objective $\mathbb{E}[\cdot]$ (would exist in reality)
 - Have restrictive existence conditions [▶ \(Details\)](#)
 - Can be tested using price data
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 - Refer to bubbles under subjective $\mathbb{E}[\cdot]$ (would only exist in investors' mind)
 - Have flexible existence conditions
 - Can only be tested with data on investors' beliefs
- So, empirical tests focus on rational bubbles
- Model-free evidence indicates no rational bubbles (Giglio, Maggiori, Stroebe (2016))



The Excess Volatility Puzzle = Time-Varying $\mathbb{E}_t[R]$

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

- The early literature considered $\mathbb{E}_t[R] = \mathbb{E}R$ a reasonable benchmark
- If $\mathbb{E}_t[R] = \mathbb{E}R$, then we must have $\text{Var}[P] < \text{Var}[P^*]$ (Shiller (1981))

$$P_t = \sum_{h=1}^{\infty} \frac{\mathbb{E}_t[CF_{t+h}]}{\mathbb{E}R^h}$$

$$P_t^* = \sum_{h=1}^{\infty} \frac{CF_{t+h}}{\mathbb{E}R^h}$$

- But $\text{Var}[P] > \text{Var}[P^*]$ in the data (this is the “Excess Volatility Puzzle”)
- An update of the evidence based on Shiller’s Nobel prize lecture (Shiller (2014))

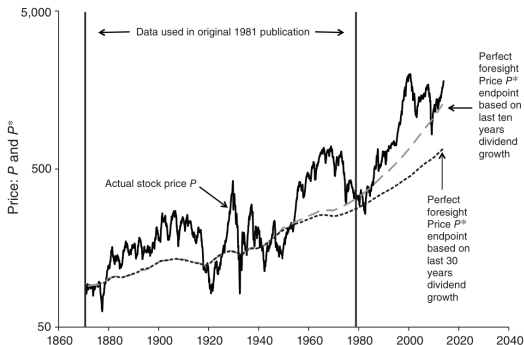


FIGURE 1. REAL STANDARD AND POOR'S COMPOSITE STOCK PRICE INDEX ALONG WITH PRESENT VALUES WITH CONSTANT DISCOUNT RATE OF SUBSEQUENT REAL DIVIDENDS ACCRUING TO THE INDEX 1871–1913.

The Excess Volatility Puzzle = Time-Varying $\mathbb{E}_t[R]$

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

- The early literature considered $\mathbb{E}_t[R] = \mathbb{E}R$ a reasonable benchmark
- If $\mathbb{E}_t[R] = \mathbb{E}R$, then we must have $\text{Var}[P] < \text{Var}[P^*]$ (Shiller (1981))

$$P_t = \sum_{h=1}^{\infty} \frac{\mathbb{E}_t[CF_{t+h}]}{\mathbb{E}R^h}$$

$$P_t^* = \sum_{h=1}^{\infty} \frac{CF_{t+h}}{\mathbb{E}R^h}$$

- But $\text{Var}[P] > \text{Var}[P^*]$ in the data (this is the “Excess Volatility Puzzle”)
- Nonstationarity: $\text{Var}[P] = \text{Var}[P^*] \rightarrow \infty$ (Kleidon (1986), Marsh, Merton (1986))
- Cointegration Theory: the P-CF deviation is stationary (Campbell, Shiller (1987))

$$P_t - \frac{CF_t}{\mathbb{E}R - 1} = \frac{\mathbb{E}R}{\mathbb{E}R - 1} \cdot \sum_{h=1}^{\infty} \frac{\mathbb{E}_t[\Delta CF_{t+h}]}{\mathbb{E}R^h}$$

- We still have $\text{Var} \left[P_t - \frac{CF_t}{\mathbb{E}R - 1} \right] > \text{Var} \left[P_t^* - \frac{CF_t}{\mathbb{E}R - 1} \right]$
- So, the Excess Volatility Puzzle is robust to stationarity issues
- And it implies $\mathbb{E}_t[R]$ varies over time

Asset Prices: CF_t vs $\mathbb{E}_t[CF \text{ Growth}]$ vs $\mathbb{E}_t[\text{Returns}]$

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{\tau=h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

- If $CF > 0$:

$$P_t = CF_t \cdot \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{\prod_{\tau=1}^{\tau=h} (CF_{t+\tau}/CF_{t+\tau-1})}{\prod_{\tau=0}^{\tau=h-1} \mathbb{E}_{t+\tau}[R]} \right]$$

$$\circ \uparrow CF_t \implies \uparrow P_t$$

$$\circ \uparrow \mathbb{E}_t[CF \text{ Growth}] \iff \uparrow P_t$$

$$\circ \uparrow \mathbb{E}_t[\text{Returns}] \iff \downarrow P_t$$

- Gordon Growth Model is a special case:

(Williams (1938), Gordon, Shapiro (1956))

$$\circ \mathbb{E}_t[CF_{t+1}/CF_t] = \mathbb{E}G \quad \& \quad \mathbb{E}_t[R] = \mathbb{E}R \quad \& \quad \mathbb{E}R > \mathbb{E}G$$

$$P_t = CF_t \cdot \sum_{h=1}^{\infty} \left(\frac{\mathbb{E}G}{\mathbb{E}R} \right)^h = CF_t \cdot \frac{\mathbb{E}G/\mathbb{E}R}{1 - \mathbb{E}G/\mathbb{E}R} = CF_t \cdot \frac{\mathbb{E}G}{\mathbb{E}R - \mathbb{E}G}$$

- Abusing the notation in the Gordon-Growth Model:

$$P_t = CF_t \cdot \frac{\mathbb{E}G_t}{\mathbb{E}R_t - \mathbb{E}G_t}$$

Campbell-Shiller Log-Linear Present Value Identity

- From the definition of a gross return with $CF > 0$ (and suppressing the j index):

$$\begin{aligned} \log(R_{t+1}) &= \log(P_{t+1} + CF_{t+1}) - \log(P_t) = \log(P_{t+1}) + \log(1 + CF_{t+1}/P_{t+1}) - \log(P_t) \\ &\downarrow \\ r_{t+1} &= p_{t+1} + \log(1 + e^{cf_{t+1} - p_{t+1}}) - p_t \end{aligned}$$

- From a first-order Taylor approximation: (Campbell, Shiller (1989))

$$p_t \approx \kappa_0 + (1 - \kappa) \cdot cf_{t+1} - r_{t+1} + \kappa \cdot p_{t+1}$$

$$w/ \quad \kappa = 1/(1 + e^{\overline{cf - P}}) \quad \& \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \cdot \log(1/\kappa - 1)$$

- Recursively substituting p_{t+h} , taking $\mathbb{E}[\cdot]$, and imposing no bubbles: $\rightarrow$ (Proof)

$$p_t \approx \frac{\kappa_0}{1 - \kappa} + (1 - \kappa) \cdot \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[cf_{t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{t+h}]$$

- The Taylor approximation can alternatively be written as (Campbell, Shiller (1989))

$$p_t - cf_t \approx \kappa_0 + \Delta cf_{t+1} - r_{t+1} + \kappa \cdot (p_{t+1} - cf_{t+1})$$

which yields (after recursively substituting $p_{t+h} - cf_{t+h}$ and taking $\mathbb{E}[\cdot]$) $\rightarrow$ (Proof)

$$p_t \approx \frac{\kappa_0 \cdot (1 - \kappa^H)}{1 - \kappa} + cf_t + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta cf_{t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{t+h}] + \underbrace{\kappa^H \cdot \mathbb{E}_t[p_{t+H} - cf_{t+H}]}_{\xrightarrow{H \rightarrow \infty} 0}$$

- We often use this log-linear present value identity (alternative: Gao, Martin (2021))

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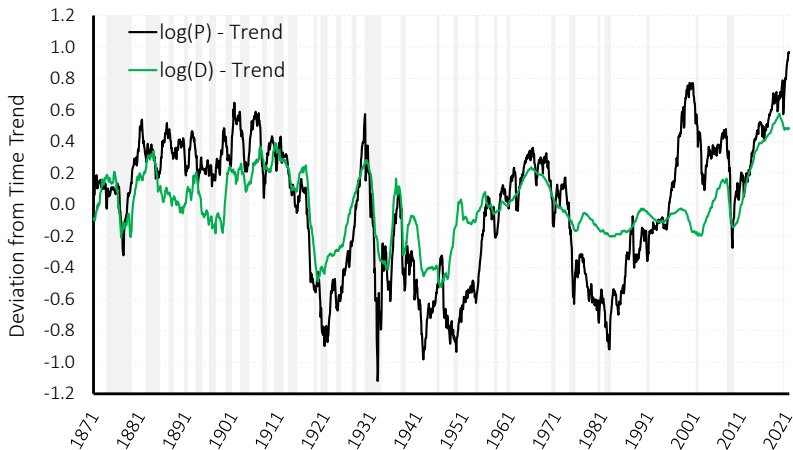
[4.2] Corporate Bonds

Source: My calculation w/ 1926-2021 CRSP data (D includes M&A and is the sum of D^m)

Equity Prices, Dividends, and the Dividend Yield

$$p_t = \frac{\kappa_0 \cdot (1 - \kappa^H)}{1 - \kappa} + d_t + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] + \kappa^H \cdot \mathbb{E}_t[p_{t+H} - d_{t+H}]$$

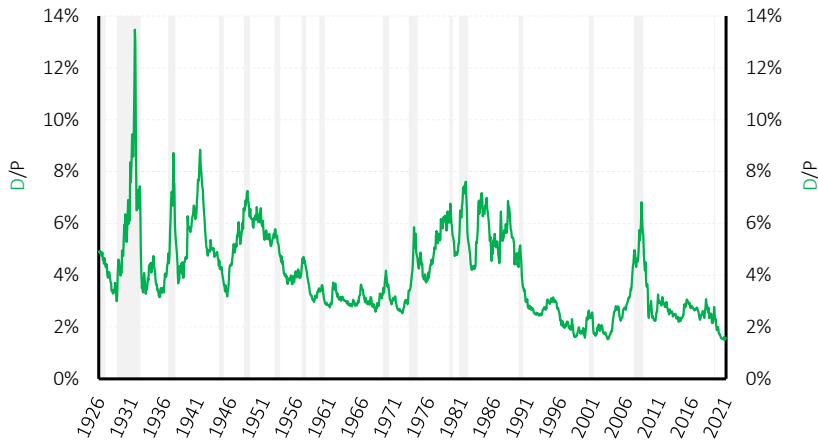
- For equities, $cf = d$ and $r = r_e$
- If $\mathbb{E}_t[\Delta d_{t+h}] = \mathbb{E}g$ and $\mathbb{E}_t[r_{e,t+h}] = \mathbb{E}r$, then:
 - p_t and d_t move in parallel
 - detrended $p_t = \text{detrended } d_t$
 - The dividend yield ($dp_t = d_t - p_t$) is constant



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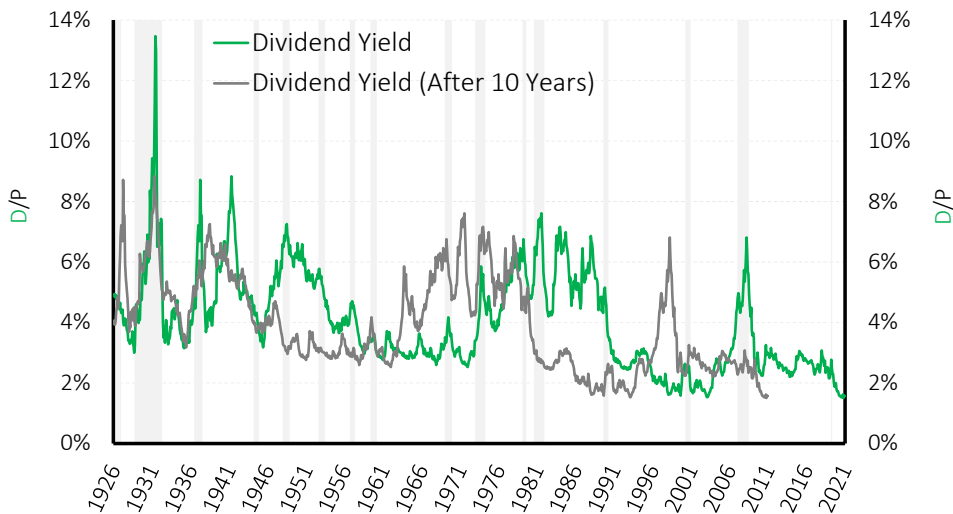
- For equities, $cf = d$ and $r = r_e$
- If $\mathbb{E}_t[\Delta d_{t+h}] = \mathbb{E}g$ and $\mathbb{E}_t[r_{e,t+h}] = \mathbb{E}r$, then:
 - p_t and d_t move in parallel
 - detrended $p_t = \text{detrended } d_t$
 - The dividend yield ($dp_t = d_t - p_t$) is constant
- Evidence indicates $\mathbb{E}_t[\Delta d_{t+h}]$ and/or $\mathbb{E}_t[r_{e,t+h}]$ are time varying
- Isolating dp_t (Campbell, Shiller (1989))

$$dp_t = \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}]$$

- $\uparrow \mathbb{E}_t[r_{e,t+h}] \iff \uparrow dp_t$
- $\uparrow \mathbb{E}_t[\Delta d_{t+h}] \iff \downarrow dp_t$
- $\uparrow \mathbb{E}_t[dp_{t+H}] \iff \uparrow dp_t$ (this mechanism $\xrightarrow{H \rightarrow \infty} 0$)

dp_t Variation: $\mathbb{E}_t[\Delta d]$ vs $\mathbb{E}_t[r_e]$

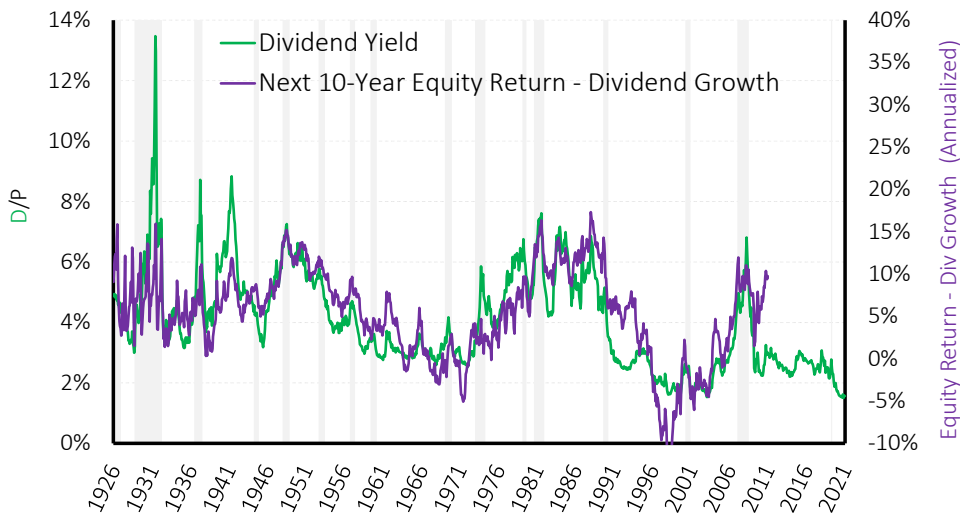
$$dp_t = \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}]$$



Source: My calculation w/ 1926-2021 CRSP data (D includes M&A and is the sum of D^m)

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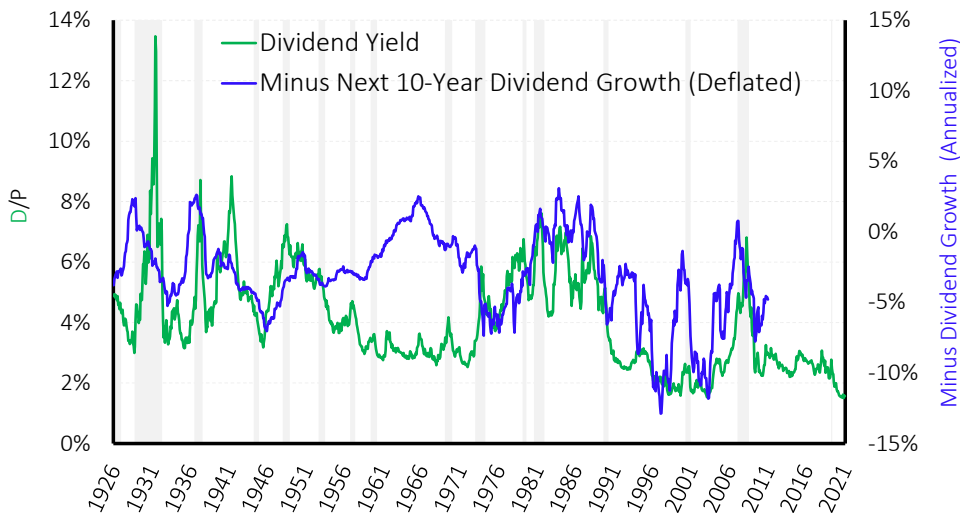
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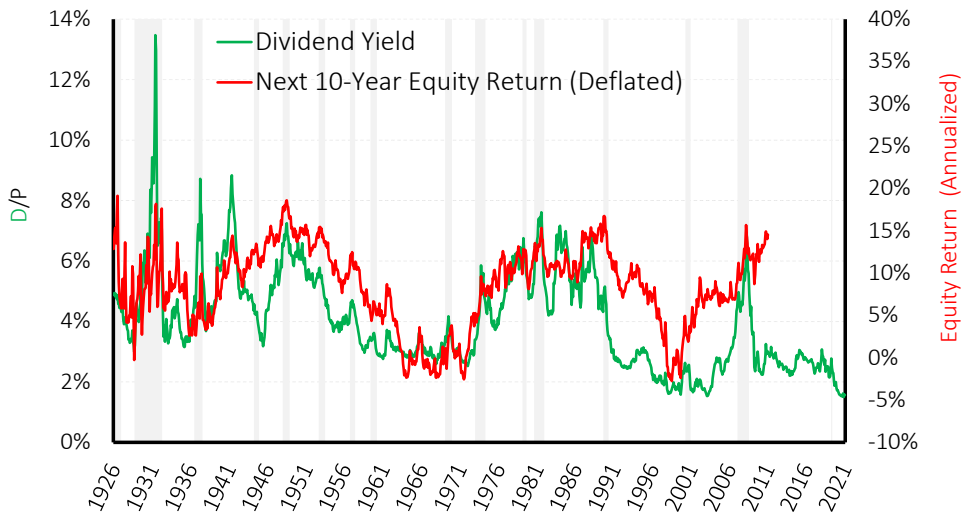
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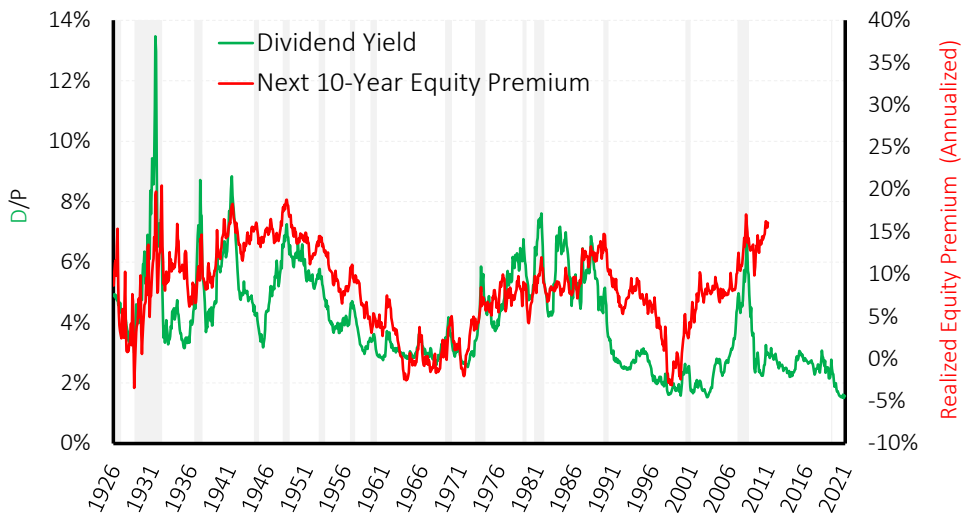
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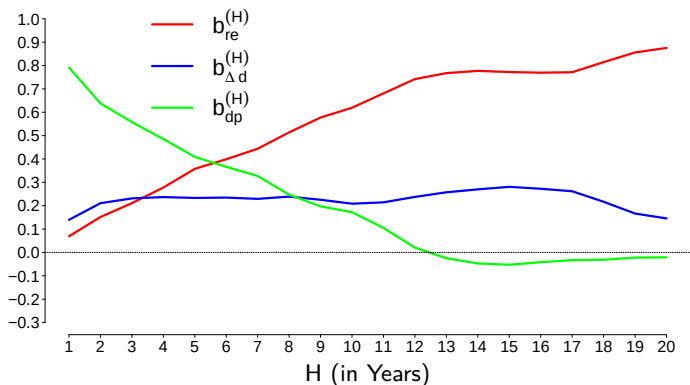
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$$dp_t = \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}]$$

- From this identity, we can formally decompose $\text{Var}[dp]$:

$$\begin{aligned} \text{Var}[dp] &= \text{Cov}\left[dp_t, \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h}\right] + \text{Cov}\left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h}\right] + \text{Cov}[dp_t, \kappa^H \cdot dp_{t+H}] \\ &\Downarrow \\ 1 &= \underbrace{\frac{\text{Cov}\left[dp_t, \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h}\right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h}\right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov}[dp_t, \kappa^H \cdot dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}} \end{aligned}$$



dp_t Variation: $\mathbb{E}_t[\Delta d]$ vs $\mathbb{E}_t[r_e]$ (through a VAR)

$$dp_t = \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}]$$

- Let $z_t = [\Delta d_t, r_{e,t}, dp_t, s_t]$ follow the first order Vector Autoregression (VAR)

$$z_{t+1} = \Gamma_0 + \Gamma z_t + \tilde{z}_{t+1} \quad \text{where} \quad \tilde{z}_{t+1} \sim IID(0, \Sigma_{z,t})$$

- In this case, $\mathbb{E}_t[z_{t+h}] = (I + \Gamma + \dots + \Gamma^{h-1})\Gamma_0 + \Gamma^h z_t$
- Letting b_z be the vector of univariate regression coefficients of z_t onto dp_t :

$$b_{re}^{(H)} = \frac{\text{Cov}\left[dp_t, \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}]\right]}{\text{Var}[dp]} = \mathbf{1}'_{re} \frac{\sum_{h=1}^H (\kappa \cdot \Gamma)^h}{\kappa} b_z = \mathbf{1}'_{re} (\Gamma - \kappa^H \cdot \Gamma^{H+1})(I - \kappa \cdot \Gamma)^{-1} b_z$$

$$b_{\Delta d}^{(H)} = -\mathbf{1}'_{\Delta d} (\Gamma - \kappa^H \cdot \Gamma^{H+1})(I - \kappa \cdot \Gamma)^{-1} b_z \quad \text{and} \quad b_{dp}^{(H)} = 1 - b_{\Delta d}^{(H)} - b_{re}^{(H)}$$

Table II Long-Run Regression Coefficients

Method and Horizon	Coefficient		
	$b_r^{(k)}$	$b_{\Delta d}^{(k)}$	$\rho^k b_{dp}^{(k)}$
Direct regression, $k = 15$	1.01	-0.11	-0.11
Implied by VAR, $k = 15$	1.05	0.27	0.22
VAR, $k = \infty$	1.35	0.35	0.00

$\tilde{p}_t$ Variation: $\tilde{d}_t$ vs $\Delta \mathbb{E}_t[\Delta d]$ vs $\Delta \mathbb{E}_t[r_e]$

$$dp_t = \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}]$$

- We can also decompose the volatility of price shocks: $\tilde{p}_{t+1} = p_{t+1} - \mathbb{E}_t[p]$
- Letting $H \rightarrow \infty$ and considering shocks:

$$\tilde{dp}_{t+1} = \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \Delta \mathbb{E}_{t+1}[r_{e,t+1+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \Delta \mathbb{E}_{t+1}[\Delta d_{t+1+h}]$$

- So that isolating $\tilde{p}_{t+1}$ yields

$$\tilde{p}_{t+1} = \tilde{d}_{t+1} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \Delta \mathbb{E}_{t+1}[\Delta d_{t+1+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \Delta \mathbb{E}_{t+1}[r_{e,t+1+h}]$$

- Substitute into Campbell-Shiller approximation, we get (Campbell (1991))

$$\tilde{r}_{e,t+1} = \underbrace{\tilde{d}_{t+1} + \sum_{h=1}^{\infty} \kappa^h \cdot \Delta \mathbb{E}_{t+1}[\Delta d_{t+1+h}]}_{N_{\Delta d, t+1}} - \underbrace{\sum_{h=1}^{\infty} \kappa^h \cdot \Delta \mathbb{E}_{t+1}[r_{e,t+1+h}]}_{N_{r_e, t+1}}$$

- $\text{Var}[dp]$ is driven by $\mathbb{E}[r_e]$ but $\text{Var}[\tilde{p}]$ and $\text{Var}[\tilde{r}_e]$ have a large $\tilde{d}_t$ component
- Cochrane (2011) “The variance of dividend yields or price-dividend ratios corresponds entirely to discount-rate variation, but as much as half of the variance of *price changes* or *returns* corresponds to current dividends. This fact seems trivial but has caused a lot of confusion”

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

- [2.1] Equity Premium Predictors
- [2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- [3.1] Equity Anomaly Signals
- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

A Log-Linear Present Value Identity for Market-to-Book

- We also want to study the importance of $\mathbb{E}_t[R]$ and $\mathbb{E}_t[CF]$ at the firm-level
- Many firms have $D_t = 0$, so we need a special log-linear present value identity
- From the definition of a gross stock return (and suppressing the j index):

$$\begin{aligned} R_{e,t+1} &= \frac{D_{t+1} + P_{t+1}}{P_t} = \frac{PO_{t+1} + ME_{t+1}}{ME_t} \\ &= \frac{BE_{t+1}}{BE_t} \cdot \left(\frac{ME_{t+1}}{BE_{t+1}} + \left(\frac{CSE_{t+1}}{BE_t} + 1 \right) \cdot \left(\frac{BE_{t+1}}{BE_t} \right)^{-1} - 1 \right) / \left(\frac{ME_t}{BE_t} \right) \end{aligned}$$

w/ $CSE_t = PO_t + \Delta BE_t$ represents clean surplus earnings

- Taking logs: $r_{e,t+1} = \Delta be_{t+1} + \log(e^{mb_{t+1}} + e^{prof_{t+1} - \Delta be_{t+1}} - 1) - mb_t$

w/ $mb_t = \log(ME_t/BE_t)$ & $prof_t = \log(1 + CSE_t/BE_{t-1})$

- From a Taylor approximation to the $\log(\cdot)$ expression (Gonçalves, Leonard (2023))

$$mb_t \approx \varphi_0 + \varpi \cdot prof_{t+1} + (1 - \varpi) \cdot \Delta be_{t+1} - r_{e,t+1} + \varphi \cdot mb_{t+1}$$

$\vdots$ recursively substituting mb_{t+h}

$$mb_t = \frac{\varphi \cdot \varphi_0}{1 - \varphi} + \sum_{h=1}^H \varphi^{h-1} \cdot (\varpi \cdot prof_{t+h} + (1 - \varpi) \cdot \Delta be_{t+h}) - \sum_{h=1}^H \varphi^{h-1} \cdot r_{e,t+h} + \varphi^H \cdot mb_{t+H}$$

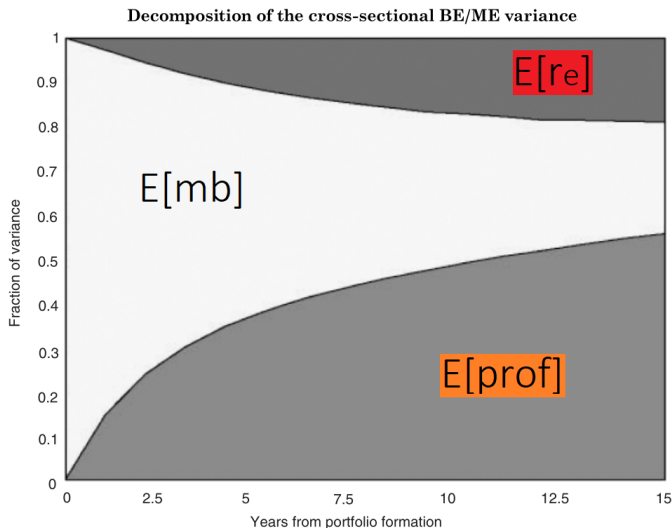
w/ $\varphi_0, \varphi, \varpi$ are functions of $\mathbb{E}[mb]$ and $\mathbb{E}[prof - \Delta be]$

Variation in ME/BE Across Firms: $\mathbb{E}_t[\text{prof}]$ vs $\mathbb{E}_t[r_e]$

$$mb_t = \frac{\varphi \cdot \varphi_0}{1 - \varphi} + \sum_{h=1}^H \varphi^{h-1} \cdot (\varpi \cdot \text{prof}_{t+h} + (1 - \varpi) \cdot \Delta be_{t+h}) - \sum_{h=1}^H \varphi^{h-1} \cdot r_{e,t+h} + \varphi^H \cdot mb_{t+H}$$

- Assuming $\mathbb{E}[mb] = 1$ (so $\varpi = 1$) and taking $\mathbb{E}[\cdot]$: (Cohen, Polk, Vuolteenaho (2003))

$$mb_t = \frac{\varphi \cdot \varphi_0}{1 - \varphi} + \sum_{h=1}^H \varphi^{h-1} \cdot \mathbb{E}_t[\text{prof}_{t+h}] - \sum_{h=1}^H \varphi^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] + \varphi^H \cdot \mathbb{E}_t[mb_{t+H}]$$



Variation in Returns over Time: $\mathbb{E}_t[\text{prof}]$ vs $\mathbb{E}_t[r_e]$

$$mb_t = \frac{\varphi \cdot \varphi_0}{1 - \varphi} + \sum_{h=1}^H \varphi^{h-1} \cdot (\varpi \cdot \text{prof}_{t+h} + (1 - \varpi) \cdot \Delta be_{t+h}) - \sum_{h=1}^H \varphi^{h-1} \cdot r_{e,t+h} + \varphi^H \cdot mb_{t+H}$$

- Take $\mathbb{E}[\cdot]$ and $H \rightarrow \infty$
- Then, use $\tilde{r}_{e,t+1} = \varpi \cdot \widetilde{\text{prof}}_{t+1} + (1 - \varpi) \cdot \widetilde{\Delta be}_{t+1} + \varphi \cdot \widetilde{mb}_{t+1}$ to get:

$$\tilde{r}_{e,t+1} = \sum_{h=0}^{\infty} \varphi^h \cdot \Delta \mathbb{E}_{t+1}[\varpi \cdot \text{prof}_{t+1+h} + (1 - \varpi) \cdot \Delta be_{t+1+h}] - \sum_{h=1}^{\infty} \varphi^h \cdot \Delta \mathbb{E}_{t+1}[r_{e,t+1+h}]$$

- Assuming $\mathbb{E}[mb] = 1$ (so $\varpi = 1$), we have (Vuolteenaho (2002))

$$\tilde{r}_{e,t+1} = \sum_{h=0}^{\infty} \varphi^h \cdot \Delta \mathbb{E}_{t+1}[\text{prof}_{t+1+h}] - \sum_{h=1}^{\infty} \varphi^h \cdot \Delta \mathbb{E}_{t+1}[r_{e,t+1+h}]$$

- Firm-level $\text{Var}[\tilde{r}_e]$ is mostly driven by variation in $\mathbb{E}_t[\text{prof}]$, not in $\mathbb{E}_t[r_e]$
- A lot of variation in $\mathbb{E}_t[\text{prof}]$ is firm-specific, diversifying at the aggregate

	Short VAR		Long VAR	
	Firm-Level	EW Index	Firm-Level	EW Index
$\% \text{Var}[\tilde{r}_e]$ from N_{re}	25%	54%	27%	54%
$\% \text{Var}[\tilde{r}_e]$ from $\widetilde{\text{prof}} + N_{\text{prof}}$	75%	46%	73%	46%

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Sample and Specification Issues (Extra)

- Stylized fact: $\text{Var}[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ (instead of $\mathbb{E}_t[\Delta d]$)

(1) Sample period matters for the predictability of Δd_{t+h}

(Chen (2009), Golez, Koudijs (2018))

- $r_{e,t+h}$ is predictable both before and after 1945
- Δd_{t+h} is predictable before 1945 but not after
- $\text{Var}[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ after 1945
 (but before 1945, both $\mathbb{E}_t[r]$ and $\mathbb{E}_t[\Delta d]$ are relevant and $\mathbb{E}_t[\Delta d]$ dominates)

Table 4 Long-horizon predictability.

	1629–1945	1945–2015	1700–2015	1629–2015
	(1)	(2)	(3)	(4)
Dependent variable: 5-year ret				
dp_t	0.44	0.44	0.34	0.32
t -statistic (Overlap.)	(4.51)	(5.25)	(4.02)	(4.29)
t -statistic (Non-overlap.)	[3.46]	[3.70]	[2.99]	[3.17]
R^2	0.15	0.28	0.13	0.12
Dependent variable: 5-year dg				
dp_t	−0.35	0.00	−0.15	−0.20
t -statistic (Overlap.)	(−3.58)	(−0.04)	(−2.48)	(−3.12)
t -statistic (Non-overlap.)	[−2.60]	[−0.04]	[−1.63]	[−2.23]
R^2	0.12	0.00	0.05	0.08

Source: Golez, Koudijs (2018)

Sample and Specification Issues (Extra)

- Stylized fact: $\text{Var}[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ (instead of $\mathbb{E}_t[\Delta d]$)

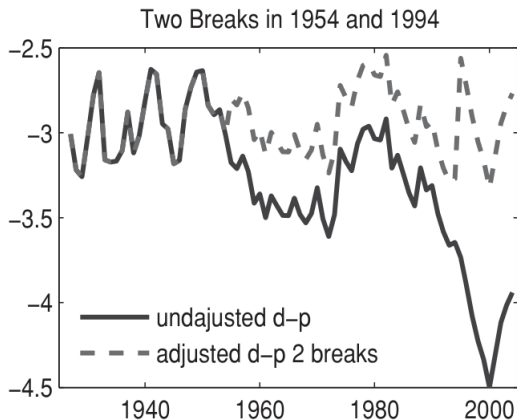
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(2) Accounting for structural breaks in the dp_t is important

(Lettau, Nieuwerburgh (2008))



Sample and Specification Issues (Extra)

- Stylized fact: $\text{Var}[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ (instead of $\mathbb{E}_t[\Delta d]$)

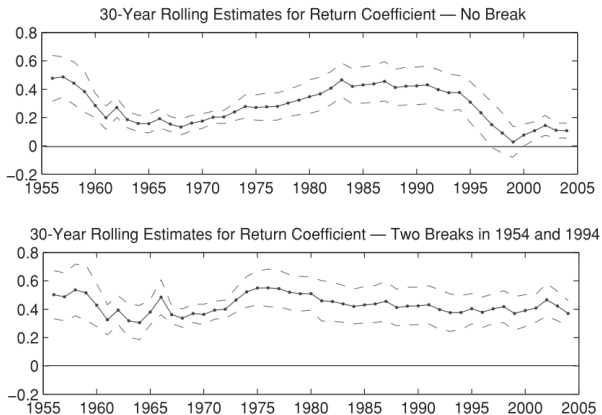
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Sample and Specification Issues (Extra)

- Stylized fact: $\mathbb{V}ar[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ (instead of $\mathbb{E}_t[\Delta d]$)

(1) Sample period matters for the predictability of Δd_{t+h}

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(but before 1945, both $\mathbb{E}_t[r]$ and $\mathbb{E}_t[\Delta d]$ are relevant and $\mathbb{E}_t[\Delta d]$ dominates)

(2) Accounting for structural breaks in the dp_t is important

(Lettau, Nieuwerburgh (2008))

(3) Choice of state vector matters when using a VAR (Chen, Zhao (2009))

Excess Return + Term Spread + Value Spread	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
10-Year PE Ratio	+							
1-Year PE Ratio			+					+
Dividend Yield				+				
Book-to-Market Ratio					+			+
Book-to-Market Spread						+	+	
Stock Variance								+
Corporate Issue							+	+
$\% \mathbb{V}ar[\tilde{r}_e]$ from N_{re}	69%	9%	57%	80%	35%	5%	11%	52%
$\% \mathbb{V}ar[\tilde{r}_e]$ from $\tilde{\Delta d} + N_{\Delta d}$	31%	91%	43%	20%	65%	95%	89%	48%

Source: Chen, Zhao (2009), implied from Table 2 (annual returns, 1952-2001)

Sample and Specification Issues (Extra)

- Stylized fact: $\text{Var}[dp]$ reflects mostly movements in $\mathbb{E}_t[r]$ (instead of $\mathbb{E}_t[\Delta d]$)

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(2) Accounting for structural breaks in the dp_t is important

(Lettau, Nieuwerburgh (2008))

(3) Choice of state vector matters when using a VAR (Chen, Zhao (2009))

- When decomposing $\text{Var}[dp]$, use direct regressions to confirm results
- When measuring $\mathbb{E}_t[r]$ and $\mathbb{E}_t[\Delta d]$ using a VAR, state vector has to include dp_t
 (Engsted et al (2012))

$$\begin{aligned}
 r_{t+1} &\approx \kappa_0 + \Delta d_{t+1} - \kappa \cdot dp_{t+1} + dp_t \\
 &\Downarrow \\
 \mathbb{E}_t[r] &\approx \kappa_0 + \mathbb{E}_t[\Delta d] - \kappa \cdot \mathbb{E}_t[dp] + dp_t \\
 &\Downarrow \\
 \mathbb{E}_t[r] - \mathbb{E}_t[\Delta d] &\approx \kappa_0 + dp_t - \kappa \cdot \mathbb{E}_t[dp]
 \end{aligned}$$

- Still, VAR results can be sensitive to the state vector used
- Solution: use a Big VAR that includes hundreds of variables (Gonçalves (2023))

Dividend Measurement Issues (Extra)

- Traditional CRSP measure of monthly dividends: $D_t^m = (R_{e,t} - R_{e,t}^{\text{ex}}) \cdot P_{t-1}$

(1) To deal with seasonality, D_t is typically annual (three versions):

(a) Reinvest D_t^m at the $R_{e,t}$ (traditional approach)

(b) Reinvest D_t^m at risk-free rate

(c) No reinvestment so that $D_t = \sum_{\tau=t-11}^{\tau=t} D_\tau^m$

- * My view: (b) and (c) are better as they do not add properties of $R_{e,t}$ into D_t
(Chen (2009), Binsbergen, Koijen (2010), Koijen, Nieuwerburgh (2011))

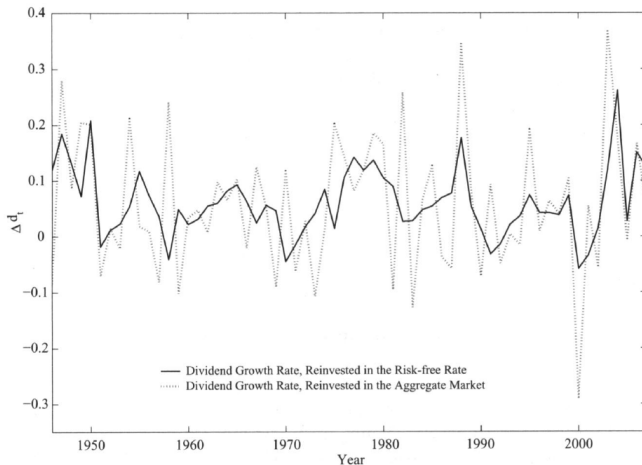


Figure 1. Dividend-growth rates: reinvesting in the risk-free rate or in the market.

Dividend Measurement Issues (Extra)

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Table 1 Return and dividend growth predictability by the dividend-price ratio^a

		Panel A: Return predictability					
		Div. reinv. at R^f			Div. reinv. at R^m		
		κ_r	t -stat	R^2	κ_r	t -stat	R^2
1	Benchmark 1926–2009	0.077	1.31	2.90	0.104	2.08	4.82
2	No approximation	0.075	1.29	2.79	0.102	2.03	4.58
3	Excess returns	0.087	1.51	3.64	0.115	2.30	5.73
4	Real returns	0.085	1.57	3.46	0.105	2.10	4.86
5	No NASDAQ	0.084	1.35	2.92	0.120	2.33	5.24
6	Benchmark 1945–2009	0.130	2.56	10.84	0.126	2.58	10.02
		Panel B: Dividend growth predictability					
		Div. reinv. at R^f			Div. reinv. at R^m		
		κ_d	t -stat	R^2	κ_d	t -stat	R^2
7	Benchmark 1926–2009	−0.078	−1.48	7.64	0.008	0.20	0.05
8	Real div. growth	−0.070	−1.63	7.61	0.009	0.23	0.07
9	No NASDAQ	−0.100	−1.74	10.55	0.008	0.22	0.05
10	Benchmark 1945–2009	0.017	0.68	1.13	0.044	1.10	2.03

Dividend Measurement Issues (Extra)

- Traditional CRSP measure of monthly dividends: $D_t^m = (R_{e,t} - R_{e,t}^{ex}) \cdot P_{t-1}$

(1) To deal with seasonality, D_t is typically annual (three versions):

(a) Reinvest D_t^m at the $R_{e,t}$ (traditional approach)

(b) Reinvest D_t^m at risk-free rate

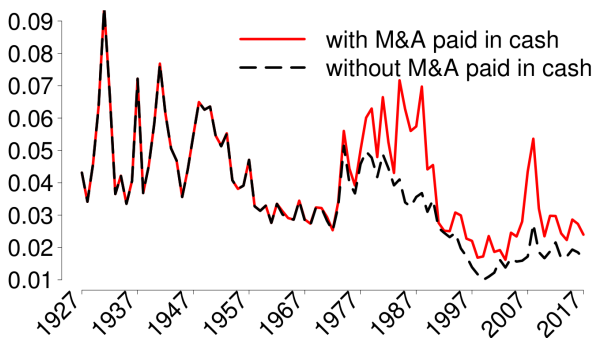
(c) No reinvestment so that $D_t = \sum_{\tau=t-11}^{\tau=t} D_\tau^m$

* My view: (b) and (c) are better as they do not add properties of $R_{e,t}$ into D_t
(Chen (2009), Binsbergen, Koijen (2010), Koijen, Nieuwerburgh (2011))

(2) D_t^m does not add M&A paid in cash (Allen, Michaely (2003), Sabbatucci (2022))

o Solution: $\hat{D}_t^m = (R_{e,t} - \hat{R}_{e,t}^{ex}) \cdot \hat{P}_{t-1}$ with $\hat{R}_{e,t}^{ex} = R_{e,t}^{ex} - M\&A \text{ yield}$

Figure IA.1. Dividend yield with and without M&A



$\mathbb{V}ar[X_t/P_t]$ Decomposition is Sensitive to X_t (Extra)

$$\begin{aligned}
 p_t &\approx \frac{\kappa_0}{1 - \kappa} + d_t + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] \\
 &\Downarrow \\
 \log(X_t/P_t) &\approx \frac{-\kappa_0}{1 - \kappa} + \log(X_t/D_t) + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}]
 \end{aligned}$$

- The literature tends to use $X_t = D_t$ because D_t and P_t are cointegrated (with the standard finding that discount rate variation explains X_t/P_t)
- But (technically) any X_t cointegrated with P_t (and thus with D_t) can be used
- In this case, X_t/D_t is treated as part of the “cash flow component”
- The aggregate literature has explored some alternative X_t beyond D_t :
 - If $X_t = BE_t$, then discount rate variation drives X_t/P_t (Vuolteenaho (2000))
 - If $X_t = E_t$, then discount rate variation drives X_t/P_t (Maio, Xu (2020))
 - If $X_t = D_t + \text{Repurchases}_t - \text{Issuances}_t$, then cash flow variation drives X_t/P_t (Larrain, Yogo (2008))
- The same issue affects the evidence on the variation of price ratios across firms:
 - If $X_t = BE_t$, then cash flow variation dominates (Cohen, Polk, Vuolteenaho (2003))
 - If $X_t = E_t$, then discount rate variation dominates (De la O, Han, Myers (2024))
- Hard to interpret the empirical evidence on price ratio variance decompositions
- Reason: $\mathbb{V}ar[X_t/P_t] \neq$ transitory price variation (see next slide)

$\mathbb{V}ar[X_t/P_t] \neq$ Transitory Price Variation (Extra)

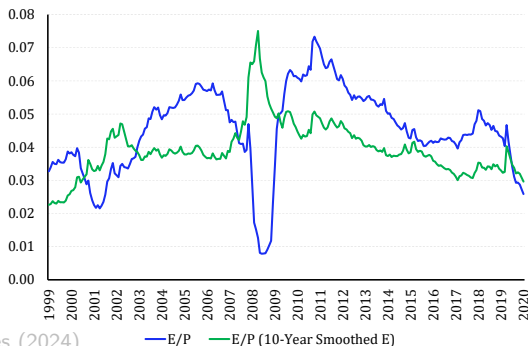
- Empirical Fact: decomposition of $\mathbb{V}ar[\log(X_t/P_t)]$ is sensitive to choice of X_t
- Econometrically, this should not be surprising (Nagel (2024))
 - If $X_t = D_t$, the fraction of $\log(X_t/P_t)$ variability from discount rates is

$$B_{Er}^D = \frac{\text{Cov} \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}], d_t - p_t \right]}{\mathbb{V}ar[d_t - p_t]}$$

- Consider an X_t that adds noise to D_t (i.e., $x_t = d_t + \varepsilon_t$). Then

$$B_{Er}^X = \frac{\text{Cov} \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}], d_t - p_t + \varepsilon_t \right]}{\mathbb{V}ar[d_t - p_t + \varepsilon_t]} = B_{Er}^D \cdot \frac{\mathbb{V}ar[d_t - p_t]}{\mathbb{V}ar[d_t - p_t] + \mathbb{V}ar[\varepsilon_t]} < B_{Er}^D$$

- Economically, the issue is that P_t/X_t dynamics may not reflect price dynamics



$\mathbb{V}ar[X_t/P_t] \neq$ Transitory Price Variation (Extra)

- Empirical Fact: decomposition of $\mathbb{V}ar[\log(X_t/P_t)]$ is sensitive to choice of X_t
- Econometrically, this should not be surprising (Nagel (2024))
 - If $X_t = D_t$, the fraction of $\log(X_t/P_t)$ variability from discount rates is

$$B_{\mathbb{E}r}^D = \frac{\text{Cov} \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}], d_t - p_t \right]}{\mathbb{V}ar[d_t - p_t]}$$

- Consider an X_t that adds noise to D_t (i.e., $x_t = d_t + \varepsilon_t$). Then

$$B_{\mathbb{E}r}^X = \frac{\text{Cov} \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}], d_t - p_t + \varepsilon_t \right]}{\mathbb{V}ar[d_t - p_t + \varepsilon_t]} = B_{\mathbb{E}r}^D \cdot \frac{\mathbb{V}ar[d_t - p_t]}{\mathbb{V}ar[d_t - p_t] + \mathbb{V}ar[\varepsilon_t]} < B_{\mathbb{E}r}^D$$

- Economically, the issue is that P_t/X_t dynamics may not reflect price dynamics
- Solution (sort of): focus on $\mathbb{V}ar \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] \right]$ (Nagel (2024))
 - We know that if $\mathbb{E}_t[r] = \mathbb{E}[r]$, then returns today are “permanent”
 - So, any variation in prices that leads to “transitory returns” comes from $\mathbb{E}_t[r]$
 - Hence, $\mathbb{V}ar \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] \right]$ is itself an interesting object

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

- [2.1] Equity Premium Predictors
- [2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- [3.1] Equity Anomaly Signals
- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

(2) Equity $\mathbb{E}[R]$ (Time Series)

- We saw that most aggregate equity variation comes from $\mathbb{E}_t[R_e - R_f] = \mathbb{E}_t[xR_e]$
- We now establish the $\mathbb{E}_t[xR_e]$ properties:
 - $\mathbb{E}_t[xR_e]$ is countercyclical (high in recessions) with $\sigma[\mathbb{E}_t[xR_e]] \approx \mathbb{E}[xR_e]$
 - Valuation ratios dominate the $\mathbb{E}_t[xR_e]$ variation at long horizons
 - Other predictors dominate the $\mathbb{E}_t[xR_e]$ variation at short and mid horizons
- To establish these properties, we use predictive regressions:

$$xR_{e,t+1} = \mathbb{E}_t[xR_e] + \epsilon_{t+1} = a + b' s_t + \epsilon_{t+1}$$

[2.1] Equity Premium Predictors

- Valuation Identity principle
- Business Cycle principle
- Risk Aversion principle
- Risk Quantity principle
- Firm Value Maximization principle
- Sentiment principle ... and others

[2.2] Important Issues and Developments

- Highlight econometric issues with return predictability
- Highlight contrast between In-Sample (IS) vs Out-of-Sample (OS) estimation
- Consider the economic significance of return predictability

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Valuation Identity Principle

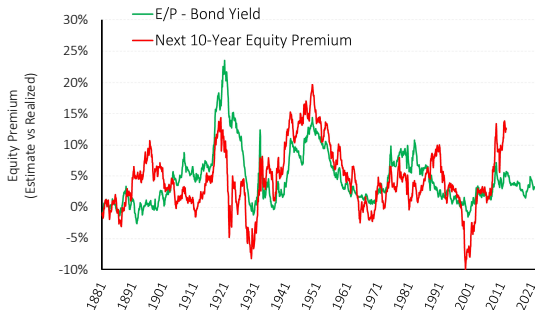
- Valuation Identity Principle:

$$p_t \approx \frac{\kappa_0}{1 - \kappa} + d_t + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}]$$

$$\Downarrow$$

$$\log(X_t/P_t) \approx \frac{-\kappa_0}{1 - \kappa} + \log(X_t/D_t) + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}]$$

- The literature has proposed several X_t/P_t as equity premium predictors:
 - Dividend Yield (Fama, French (1988), Campbell, Shiller (1989))
 - Earnings Yield (smoothed) (Campbell, Shiller (1988), Hillenbrand, McCarthy (2025))
 - Equity Payout Yield (Boudoukh et al (2007))
 - Book-to-Market Ratio (Kothari, Shanken (1997), Pontiff, Schall (1998))
- Equity valuation ratios tend to predict well long-run returns (horizon > 5 years)



Valuation Identity Principle

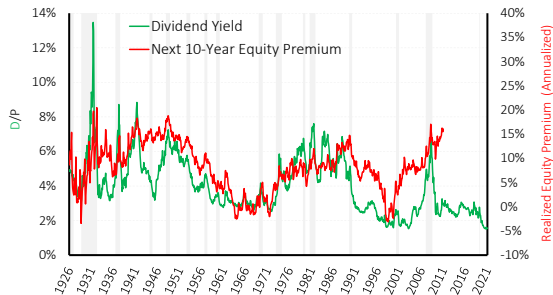
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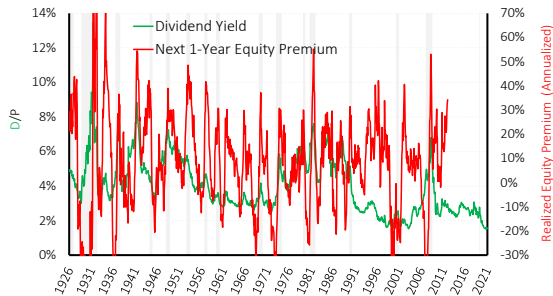
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$$\log(X_t/P_t) \approx \frac{-\kappa_0}{1 - \kappa} + \log(X_t/D_t) + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}]$$

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$$\begin{aligned}
 p_t &\approx \frac{\kappa_0}{1-\kappa} + d_t + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] \\
 &\quad \Downarrow \\
 \log(X_t/P_t) &\approx \frac{-\kappa_0}{1-\kappa} + \log(X_t/D_t) + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}]
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Table I Return-Forecasting Regressions

The regression equation is $R_{t \rightarrow t+k}^e = a + b \times D_t/P_t + \varepsilon_{t+k}$. The dependent variable $R_{t \rightarrow t+k}^e$ is the CRSP value-weighted return less the 3-month Treasury bill return. Data are annual, 1947–2009. The 5-year regression t -statistic uses the Hansen–Hodrick (1980) correction. $\sigma[E_t(R^e)]$ represents the standard deviation of the fitted value, $\sigma(b \times D_t/P_t)$.

Horizon k	b	$t(b)$	R^2	$\sigma[E_t(R^e)]$	$\frac{\sigma[E_t(R^e)]}{E(R^e)}$
1 year	3.8	(2.6)	0.09	5.46	0.76
5 years	20.6	(3.4)	0.28	29.3	0.62

Source: Cochrane (2011)

Valuation Identity Principle (Consumption-Wealth Ratio)

- Household's budget constraint is $W_{t+1} = (W_t - C_t) \cdot R_{w,t+1}$
- Log-linearize this budget constraint to get (Lettau, Ludvigson (2001))

$$\log(C_t/W_t) = cw_t \approx const + \sum_{h=1}^{\infty} \kappa_w^{h-1} \cdot \mathbb{E}_t[r_{w,t+h}] - \sum_{h=1}^{\infty} \kappa_w^{h-1} \cdot \mathbb{E}_t[\Delta C_{t+h}]$$

- Analogous to the valuation identity of the previous slide, cw_t must predict:
 - Future log returns on the wealth portfolio (r_w) or
 - Future log consumption growth (ΔC)
- We observe financial wealth (A_t) but not labor wealth (H_t): cannot measure cw_t
- Solution: adjust the cw_t equation (Lettau, Ludvigson (2001))
 - Approximate $W_t = A_t + H_t$ as $w_t \approx \omega \cdot a_t + (1 - \omega) \cdot h_t$
 - Use $h_t = y_t - yh_t$, where $yh_t = \log(Y_t/H_t)$ is the labor income-to-wealth ratio
 - Substituting w_t and h_t into the cw_t equation:

$$cay_t \approx const + \sum_{h=1}^{\infty} \kappa_w^{h-1} \cdot \mathbb{E}_t[r_{w,t+h}] - \sum_{h=1}^{\infty} \kappa_w^{h-1} \cdot \mathbb{E}_t[\Delta C_{t+h}] - (1 - \omega) \cdot yh_t$$

where $cay_t = C_t - \omega \cdot a_t - (1 - \omega) \cdot y_t$ is the cointegrating residual

- Note that if yh_t is not too volatile, then cay_t is a good proxy for cw_t
- Estimate ω by Dynamic OLS (Brennan, Xia (2005) vs Lettau, Ludvigson (2005))

Valuation Identity Principle (Consumption-Wealth Ratio)

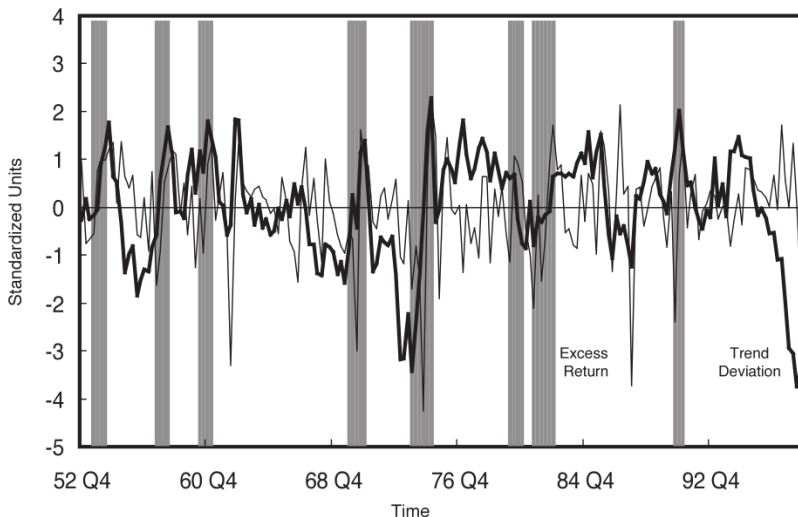


Figure 1. Excess returns and trend deviations. Excess return is the return on the S&P Composite Index less the return on the three-month Treasury bill rate. Trend deviation is the estimated deviation from the shared trend in consumption c , labor income y , and asset wealth a : $\bar{c}\bar{a}\bar{y}_t = c_t - \hat{\beta}_a a_t - \hat{\beta}_y y_t$. Both series are normalized to standard deviations of unity. The sample period is fourth quarter of 1952 to third quarter of 1998. Shaded areas denote NBER recessions.

Source: Lettau, Ludvigson (2001)

Valuation Identity Principle (Business Cycle vs Long-Run)

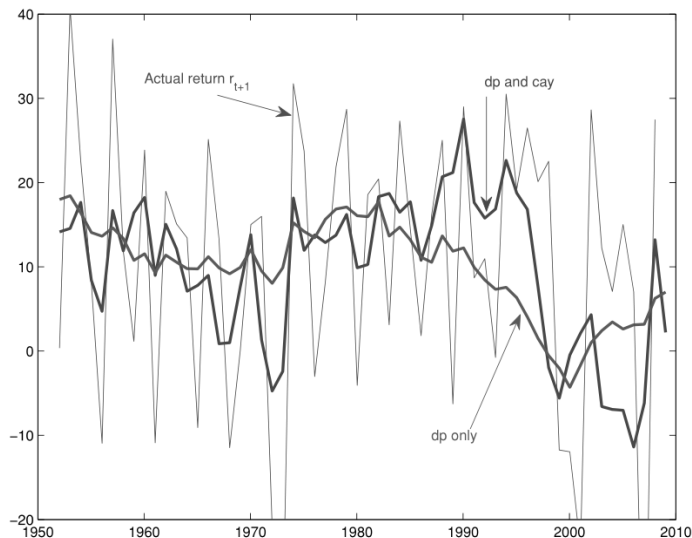


Figure 3. Forecast and actual 1-year returns. The forecasts are fitted values of regressions of returns on dividend yield and *cay*. Actual returns r_{t+1} are plotted on the same date as their forecast, $a + b \times dp_t$.

Source: Cochrane (2011)

- *cay* captures expected returns at the business cycle frequency
- *dp* captures expected long-run returns
- *cay* has almost no predictive power (beyond *dp*) for long-run returns

Valuation Identity Principle (Business Cycle vs Long-Run)

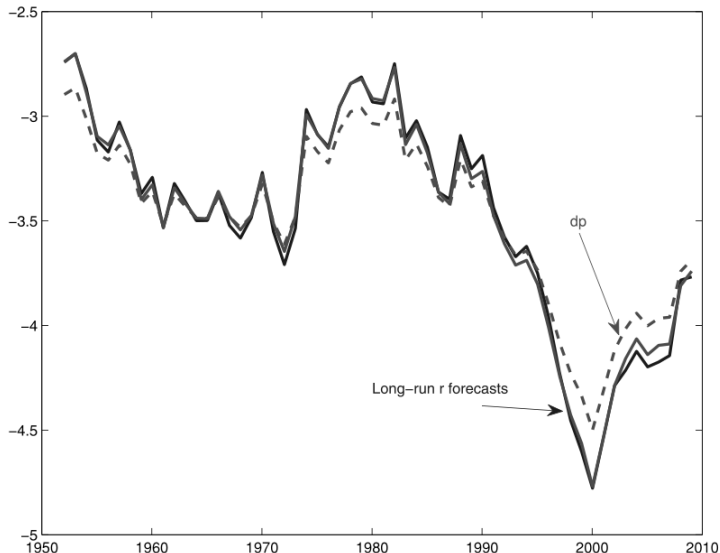


Figure 4. Log dividend yield dp and forecasts of long-run returns $\sum_{j=1}^{\infty} \rho^{j-1} r_{t+j}$. Return forecasts are computed from a VAR including dp , and a VAR including dp and cay .

Source: Cochrane (2011)

- cay captures expected returns at the business cycle frequency
- dp captures expected long-run returns
- cay has almost no predictive power (beyond dp) for long-run returns

Business Cycle Principle

- Business Cycle Principle:
 - In general equilibrium, investment opportunities vary over the business cycle
 - As such, variables that predict business cycles are also equity premium predictors
- Several equity premium predictors capture this logic:
 - Short-Term Interest Rates (Fama, Schwert (1977), Fama (1981))
 - Interest Rate Term Spread (Campbell (1987) and Fama, French (1989))
 - Corporate Default Yield Spread (Keim, Stambaugh (1986))
 - Credit Conditions (Chava, Gallmeyer, Park (2015))
 - Output Gap (Cooper, Priestley (2009))
 - Business Cycle Turning Points (Gómez-Cram (2022))

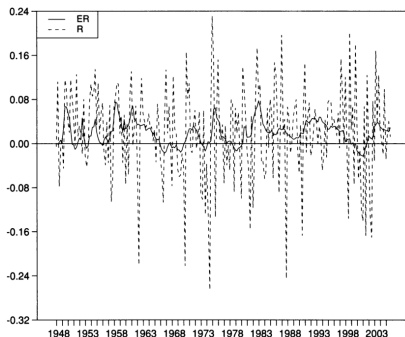
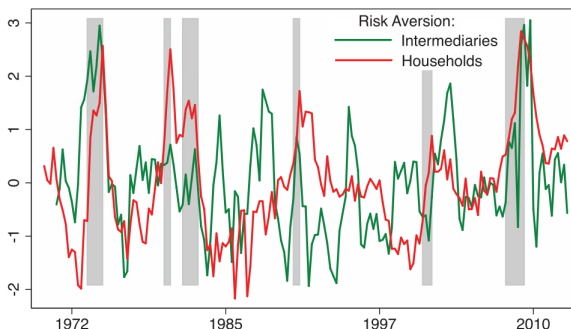


Figure 2
Actual returns (R) and expected returns (ER)

Risk Aversion Principle

- Risk Aversion Principle:
 - Some models imply that periods of high $\partial U(\cdot)$ lead to high risk aversion (γ_t)
 - When γ_t is high, investors do not want to take risk, driving up the equity premium
- Several equity premium predictors capture this logic:
 - Surplus Consumption Ratio (Wachter (2006), Ghattassi (2008))
 - Cyclical Consumption (Atanasov, Møller, Priestley (2021))
 - Unemployment Gap (Atanasov (2015))
 - Intermediary Capital Ratio (He, Kelly, Manela (2017))
 - Tightness of Intermediary Constraints (Chen, Joslin, Ni (2019))
 - Intermediary Risk Aversion (Haddad, Muir (2021))



Risk Quantity Principle

- Risk Quantity Principle:
 - Investors dislike risk and uncertainty
 - In times of high risk or uncertainty, investors require high equity premium
- Several equity premium predictors capture this logic:
 - Conditional Market Volatility (Ghysels, Santa-Clara, Valkanov (2005))
 - Risk Neutral Market Volatility (Bakshi, Panayotov, Skoulakis (2011), Martin (2017))
 - News Implied Market Volatility (Manela, Moreira (2017))
 - Aggregate Stock Tail Risk (Kelly, Jiang (2014))
 - Systematic Default Probability (Bao, Hou, Zhang (2023))
 - Dispersion and Skewness in Expected GDP Growth (Colacito et al (2016))

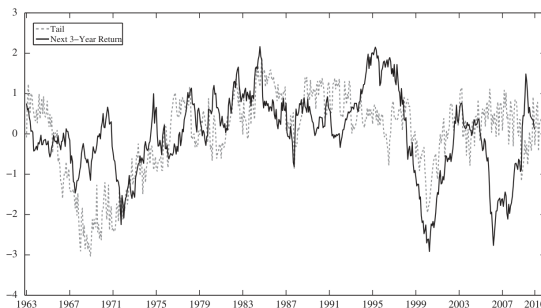


Figure 1 Tail exponent estimates and subsequent market returns

Firm Value Maximization Principle

- Firm Value Maximization Principle:
 - Managers maximize firm value while taking their cost of equity as given
 - So, firm decisions provide information about the equity premium
- Several equity premium predictors capture this logic:
 - Firm Investment Rate (Cochrane (1991), Arif, Lee (2014))
 - Firm Investment Plans (Li, Wang, Yu (2021))
 - Firm Hiring Rate (Belo et al (2023))
 - Firm Wage Growth (Favilukis, Lin (2016))
 - Firm New Orders / Shipments (Jones, Tuzel (2013))
 - Firm Net Payout over Output (Gonçalves, Stathopoulos (2025))

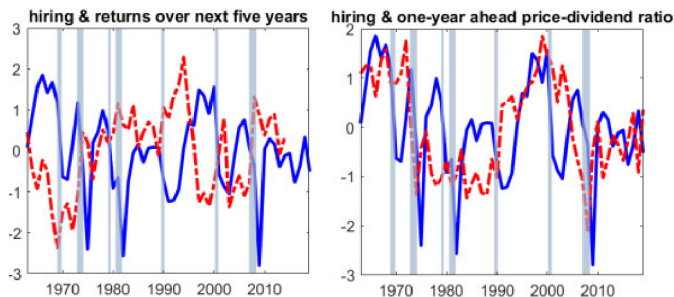
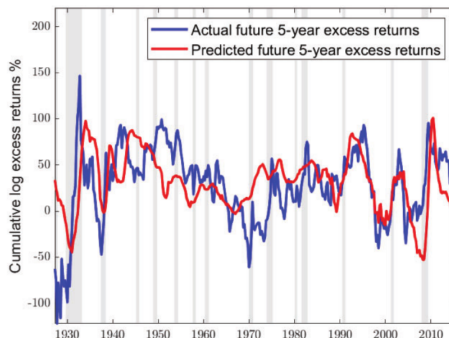


Figure 1 Time-series plots aggregate gross hiring rate (solid lines in the plots).

Sentiment Principle

- Sentiment Principle
 - Overly optimistic investors drive up prices leading to low equity premium
 - Overly pessimistic investors drive down prices leading to high equity premium
- Several equity premium predictors capture this logic:
 - Aligned Investor Sentiment (Huang et al (2015))
 - Manager Sentiment (Jiang et al (2019))
 - Employee Sentiment (Chen et al (2023))
 - Music Sentiment (Edmans et al (2022))
 - Short Interest (Rapach, Ringgenberg, Zhou (2016))
 - Experienced Payout Growth (Nagel, Xu (2022))



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Bias in Predictive Coefficient Estimates and Standard Errors

- Bias in coefficient estimates for persistent predictors: (Stambaugh (1999))

$$xR_{e,t+1} = a + b \cdot s_t + \epsilon_{t+1}$$

$$s_{t+1} = \theta + \phi \cdot s_t + u_{t+1}$$

- In small samples, $\mathbb{E}[\hat{\phi} - \phi] < 0$ (standard issue in OLS estimation of AR(1))
- Similarly, $\hat{b}$ is biased:

$$\mathbb{E}[\hat{b} - b] = \frac{\text{Cov}[\epsilon, u]}{\text{Var}[u]} \cdot \mathbb{E}[\hat{\phi} - \phi] \approx \frac{-\text{Cov}[\epsilon, u]}{\text{Var}[u]} \cdot \left(\frac{1 + 3 \cdot \phi}{T} \right)$$

- $\text{Cov}[\epsilon, u] < 0 \Rightarrow \mathbb{E}[\hat{b}] > b$ (happens with $s = \log(X/P)$)
- Correction for estimate: Lewellen (2004), Amihud, Hurvich (2004)
- Correction for inference: Campbell, Yogo (2006), Amihud, Hurvich, Wang (2009)
- Bias in inference for regressions predicting long-horizon (overlapping) returns:
 - Predicting long-horizon returns $\Rightarrow$ returns overlap $\Rightarrow$ autocorrelated residuals
 - Need to adjust standard errors (see Hansen, Hodrick (1980) and Hodrick (1992))
 - R^2 increases with horizon even if $b = 0$ (Boudoukh, Richardson, Whitelaw (2008))
 - But statistical evidence on return predictability remains strong (Cochrane (2008))

In-Sample (IS) vs Out-of-Sample (OS) Equity Premium

- Summary of empirical evidence (equity premium properties):
 - Equity premium varies over time (with $\sigma[\mathbb{E}_t[xR_e]] \approx \mathbb{E}[xR_e]$)
 - Equity premium is countercyclical (high in recessions/bad times)
 - Valuation ratios dominate the equity premium variation at long horizons
 - Other predictors dominate the equity premium variation at short and mid horizons
- We traditionally rely on “In-Sample” (IS) estimation of the equity premium:

$$\text{Model :} \quad xR_{e,t+1} = a + b \cdot s_t + \epsilon_{t+1}$$

$$\text{Estimate :} \quad \mathbb{E}_t[xR_e] = \hat{a} + \hat{b} \cdot s_t$$

- It is IS since $\hat{a}$ and $\hat{b}$ are estimated with the full sample
- In $t = 1990$, an investor would not know $\mathbb{E}_t[xR_e]$ (because of $\hat{a}$ and $\hat{b}$)
- Does $\mathbb{E}_t[xR_e]$ really reflect the “equity premium” investors required at t ?
- Goyal and Welch (2008) emphasize “Out-of-Sample” (OS) estimation:

$$\text{Model :} \quad xR_{e,t+1} = a + b \cdot s_t + \epsilon_{t+1}$$

$$\text{Estimate :} \quad \mathbb{E}_t[xR_e] = \hat{a}_t + \hat{b}_t \cdot s_t$$

- It is OS since $\hat{a}_t$ and $\hat{b}_t$ are estimated using data only up to t
- In $t = 1990$, an investor would know $\mathbb{E}_t[xR_e]$
- Easier to argue that $\mathbb{E}_t[xR_e]$ reflects the “equity premium” investors required at t

In-Sample (IS) vs Out-of-Sample (OS) Equity Premium

Table 1
Forecasts at annual frequency

Table 2
Forecasts at 5-year frequency

	Variable	Data	annual OS $\bar{R}^2$	5-year OS $\bar{R}^2$
dfy	Default yield spread	1919–2005	−3.29	−59.33
infl	Inflation	1919–2005	−4.07	−11.25
svar	Stock variance	1885–2005	−27.14	−79.33
d/e	Dividend payout ratio	1872–2005	−4.33	−4.87
lty	Long term yield	1919–2005	−7.72	−122.13
tms	Term spread	1920–2005	−2.42	−26.52
tbl	Treasury-bill rate	1920–2005	−3.37	−17.66
dfr	Default return spread	1926–2005	−2.16	−5.71
d/p	Dividend price ratio	1872–2005	−2.06	−1.19*
d/y	Dividend yield	1872–2005	−1.93	−4.45
ltr	Long term return	1926–2005	−11.79	−7.40
e/p	Earning price ratio	1872–2005	−1.78	−1.04
b/m	Book to market	1921–2005	−1.72	−13.06
i/k	Invstmnt capital ratio	1947–2005	−1.77	12.99**
ntis	Net equity expansion	1927–2005	−5.07	−3.46
eqis	Pct equity issuing	1927–2005	2.04**	−2.35
all	Kitchen sink	1927–2005	−139.03	−499.83

Source: Goyal, Welch (2008)

$$R_{OS}^2 = 1 - \frac{\sum_{t=1}^T (xr_{e,t+1} - \mathbb{E}_t[xr_e])^2}{\sum_{t=1}^T (xr_{e,t+1} - \bar{xr}_{e,t})^2}$$

$$\bar{R}_{OS}^2 = R_{OS}^2 - (1 - R_{OS}^2) \cdot \left(\frac{T - k}{T - 1} \right)$$

Responses to the IS vs OS Estimation Issue

1. IS is statistically better than OS to test for predictability (Inoue and Kilian (2007))
 - Data mining and dynamic misspecification cannot explain IS vs OS differences
 - Under the null of no predictability, IS and OS tests are asymptotically equivalent
 - Under the alternative of predictability, IS tests are statistically more powerful
2. IS is better than OS to test some economic models (Lettau, Ludvigson (2005))
 - In some models, agents know the underlying predictability parameters
 - In some models, predictability is an equilibrium outcome agents do not trade on
3. Economic restrictions improve OS predictability (Campbell, Thompson (2008))
 - Focus on the raw equity premium $\mathbb{E}_t[xR_e]$, not logged equity premium $\mathbb{E}_t[xr_e]$
 - Set $\mathbb{E}_t[xR_e] = \overline{xR_{e,t}}$ whenever $\text{sign}(\hat{b}_t) \neq \text{sign}(\hat{b})$
 - Set $\mathbb{E}_t[xR_e] = 0$ whenever $\hat{a}_t + \hat{b}_t \cdot s_t < 0$
 - Impose restrictions from steady-state valuation models ($\pi = D/E$):

$$\mathbb{E}_t[R_e] = \frac{D_t}{P_t} + (1 - \pi) \cdot \text{ROE}_t$$

$$\mathbb{E}_t[R_e] = \pi \cdot \frac{E_t}{P_t} + (1 - \pi) \cdot \text{ROE}_t \xrightarrow{\pi=\pi^*} \frac{E_t}{P_t}$$

$$\mathbb{E}_t[R_e] = \text{ROE}_t \cdot \left(1 + \pi \cdot \left(\frac{BE_t}{ME_t} - 1\right)\right)$$

Responses to the IS vs OS Estimation Issue

Table 1 Excess return prediction with regression constraints

Monthly Returns	Out-of-Sample R -squared with Different Constraints			
	Unconstrained	Positive Slope	Pos. Forecast	Both
Dividend-price ratio	-0.65%	0.05%	0.07%	0.08%
Earnings-price ratio	0.12	0.18	0.14	0.18
Smooth earnings-price ratio	0.33	0.42	0.38	0.43
Book-to-market	-0.43	-0.43	0.00	0.00
ROE	-0.93	-0.06	-0.93	-0.06
T-Bill rate	0.52	0.51	0.57	0.55
Long-term yield	-0.19	-0.19	0.20	0.20
Term spread	0.46	0.47	0.45	0.46
Default spread	-0.19	-0.19	-0.19	-0.19
Inflation	-0.22	-0.21	-0.18	-0.17
Net equity issuance	0.34	0.34	0.50	0.50
Consumption-wealth ratio	-1.36	-1.36	0.27	0.27

Source: Campbell, Thompson (2008)

Responses to the IS vs OS Estimation Issue

Table 1 Excess return prediction with regression constraints

Annual Returns	Out-of-Sample R -squared with Different Constraints			
	Unconstrained	Positive Slope	Pos. Forecast	Both
Dividend-price ratio	5.53	5.53	5.63	5.63
Earnings-price ratio	4.93	4.93	4.94	4.94
Smooth earnings-price ratio	7.89	7.89	7.85	7.85
Book-to-market	-3.38	-3.38	1.39	1.39
ROE	-8.60	-0.03	-8.35	-0.03
T-Bill rate	5.54	5.54	7.47	7.47
Long-term yield	-0.15	-0.15	2.26	2.26
Term spread	4.79	4.79	4.74	4.74
Default spread	-3.81	-3.81	-3.81	-3.81
Inflation	-0.71	-0.71	-0.71	-0.71
Net equity issuance	-4.27	-4.27	-2.38	-2.38
Consumption-wealth ratio	-7.75	-7.75	-1.48	-1.48

Source: Campbell, Thompson (2008)

$$R_{OS}^2 = 1 - \frac{\sum_{t=1}^T (xR_{e,t+1} - \mathbb{E}_t[xR])^2}{\sum_{t=1}^T (xR_{e,t+1} - \overline{xR_{e,t}})^2}$$

Responses to the IS vs OS Estimation Issue

Table 2 Excess return prediction with valuation constraints

Monthly Returns	OS R -squared	
	Unconstrained	Fixed Coefs
Dividend/price + growth	-0.05	0.63
Earnings/price + growth	-0.05	0.57
Smooth earnings/price + growth	0.11	0.72
Book-to-market + growth	-0.35	0.33
Earnings/price	0.12	0.76
Smooth earnings/price	0.32	0.97

Annual Returns	OS R -squared	
	Unconstrained	Fixed Coefs
Dividend/price + growth	2.49	4.35
Earnings/price + growth	1.69	3.89
Smooth earnings/price + growth	3.16	5.39
Book-to-market + growth	-3.53	3.63
Earnings/price	4.93	5.87
Smooth earnings/price	7.89	7.99

Economic Significance of Return Predictability (Extra)

- Even though $R_{OS}^2 > 0$, the R_{OS}^2 values look small
- We can evaluate the economic significance of R_{OS}^2 (Campbell, Thompson (2008))

- Let $xR_{e,t+1} = \mu_t + \epsilon_{t+1}$ with $\mu_t \sim N(\mu, \sigma_\mu^2)$ and $\epsilon_t \sim N(0, \sigma_\epsilon^2)$
- The investor has mean-variance utility:

$$U_t = \mathbb{E}_t[R_p] - 0.5 \cdot \gamma \cdot \text{Var}_t[R_p] = R_{f,t} + \omega_t \cdot \mu_t - 0.5 \cdot \gamma \cdot \omega_t^2 \cdot \sigma_\epsilon^2$$

- If the investor recognizes return predictability, then $\omega_t = \frac{1}{\gamma} \cdot \frac{\mu_t}{\sigma_\epsilon^2}$
- If the investor does not recognize return predictability, then $\omega = \frac{1}{\gamma} \cdot \frac{\mu}{\sigma_\mu^2 + \sigma_\epsilon^2}$
- So, the expected utility gain from recognizing return predictability is

$$\mathbb{E}[\Delta U] = \mathbb{E}[(\omega_t - \omega) \cdot \mu_t] - 0.5 \cdot \mathbb{E}[\gamma \cdot (\omega_t^2 - \omega^2) \cdot \sigma_\epsilon^2]$$

- Some algebra reveals that ▶ (Proof)

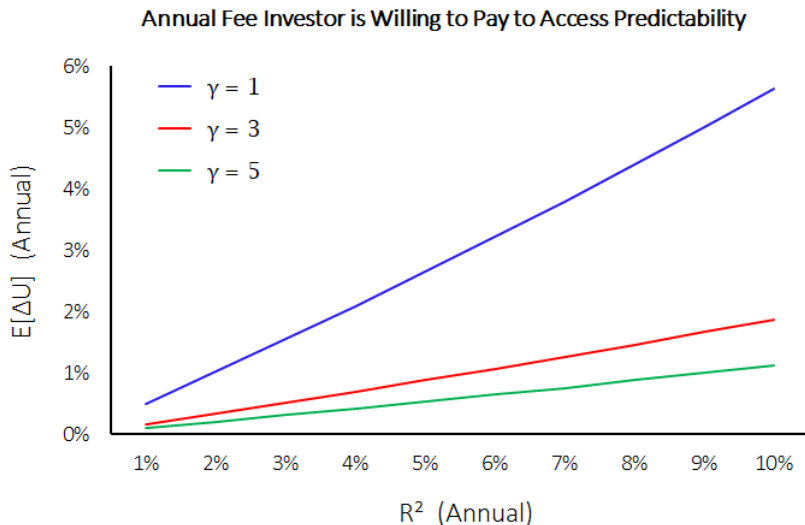
$$\mathbb{E}[(\omega_t - \omega) \cdot \mu_t] = \frac{1}{\gamma} \cdot \frac{R^2 \cdot (1 + SR^2)}{1 - R^2}$$

$$\mathbb{E}[\gamma \cdot (\omega_t^2 - \omega^2) \cdot \sigma_\epsilon^2] = \frac{1}{\gamma} \cdot \frac{R^2 + SR^2 \cdot (1 - (1 - R^2)^2)}{1 - R^2}$$

- $\mathbb{E}[\Delta U]$ depends on

- * The investor risk aversion: γ (Campbell, Thompson (2008) set $\gamma = 3$)
- * The risky asset Sharpe Ratio: SR (Campbell, Thompson (2008) set $SR = 0.374$)
- * The predictive R-squared for the risky asset: R^2
- * e.g., investor is willing to pay a $\mathbb{E}[\Delta U] = 1.1\%$ annual fee to access $R^2 = 6\%$

Economic Significance of Return Predictability (Extra)



Source: My calculation with $SR = 0.374$ following Campbell, Thompson (2008)

Advanced Forecasting Methods (Extra)

- New predictors also tend to fail out-of-sample (Goyal, Welch, Zafirov (2024))
- Predictive regressions are very simple but not the best OS forecasting method
- We can improve R_{OS}^2 by using advanced forecasting methods
 - Extract predictors using principal components (Diffusion Index Forecasting) (Ludvigson, Ng (2007))
 - Average $\mathbb{E}_t[xR]$ from univariate predictive regressions (Forecast Combinations) (Rapach, Strauss, Zhou (2010))
 - Extract $\mathbb{E}_t[xR]$ from cross-section of book-to-market ratios (Partial Least Squares) (Kelly, Pruitt (2013))
 - Extract $\mathbb{E}_t[xR]$ from cross-section of stock returns (using shrinkage techniques) (Dong et al (2022))
- Rapach, Zhou (2022) provides a review of this literature
- Complexity is theoretically and empirically useful in forecasting market returns (Kelly, Malamud, Zhou (2024))
- My view: advanced forecasting methods (may) help predict short- and mid-term returns, but valuation ratios remain the best predictors of long-term returns

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

- [2.1] Equity Premium Predictors
- [2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- [3.1] Equity Anomaly Signals
- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- We saw that a large part of cross-firm price variation comes from $\mathbb{E}_t[R_j]$
- We now link $\mathbb{E}_t[R_j]$ to firm-level characteristics (referred to as “anomaly signals”)
- To do so, we use cross-sectional regressions and portfolio sorts

[3.1] Equity Anomaly Signals

- Momentum Signals
- Friction Signals
- Value Signals
- Investment Signals
- Profitability Signals
- ...and others

[3.2] Important Issues and Developments

- Publication Incentives
 - * we collectively data mine to find and publish equity anomaly signals
- Scientific Replication
 - * typical signal $\mathbb{E}[R]$ tends to decline after reasonable changes in methodology
- Trading Costs & Short-Selling Fees
 - * typical signal has no $\mathbb{E}[R]$ post 2005 after trading costs or short-selling fees
- Anomaly Long-Term $\mathbb{E}[R]$
 - * typical signal has no $\mathbb{E}[R]$ over the long-run (little price impact)

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Testing an Anomaly Signal: Fama-MacBeth Regressions

- How can we test whether a firm-level signal ($s_{j,t}$) represents an “anomaly”?
- With the panel data regression

$$xR_{j,t+1} = a + b \cdot s_{j,t} + \theta' z_{j,t} + \epsilon_{j,t+1}$$

(1) In Standard Econometrics: Pooled OLS Regression

- Estimate the regression by OLS
- Estimate standard errors using some clustering method

(2) In Asset Pricing: Fama-MacBeth Regressions (aka Cross Sectional Regressions)

- Estimate cross-sectional regressions period by period

$$xR_{j,t+1} = \hat{a}_{t+1} + \hat{b}_{t+1} \cdot s_{j,t} + \tilde{\theta}_{t+1}' z_{j,t} + \hat{\epsilon}_{j,t+1}$$

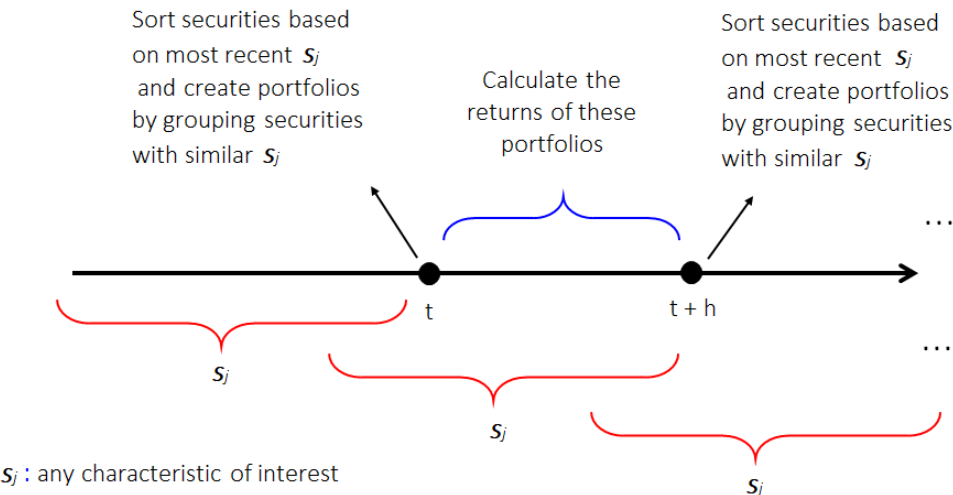
- Obtain b estimate from time-series regression (analogous for other parameters)

$$\hat{b}_t = b + \eta_t^b$$

- Standard error for the $\hat{b}$ is robust to cross-firm correlation in $\epsilon_{j,t}$
- We can also make it robust to autocorrelation in $\epsilon_{j,t}$ (Newey, West (1987, 1994))
- $s_{j,t} = s_j \Rightarrow$ Pooled OLS $\hat{b} =$ Fama-MacBeth $\hat{b}$ (Ch 12.3 in Cochrane (2005))
- $\hat{b}_t$ is the $xR_{p,t}$ on a portfolio with $s_{p,t} = 1$ and $z_{p,t} = 0$ (Balduzzi, Robotti (2008))

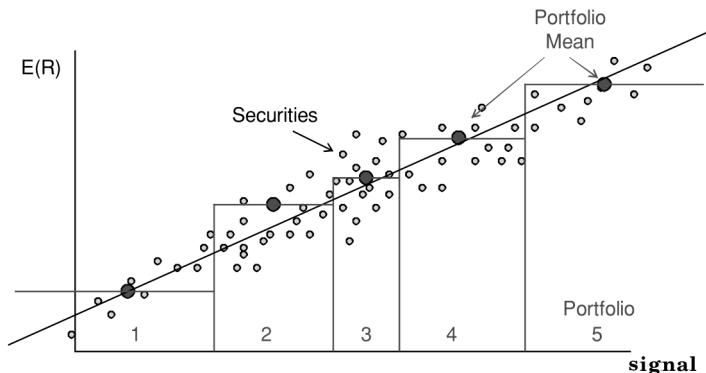
Testing an Anomaly Signal: Portfolio Sorts

- How can we test whether a firm-level signal ($s_{j,t}$) represents an “anomaly”?



Testing an Anomaly Signal: Portfolio Sorts

- How can we test whether a firm-level signal ($s_{j,t}$) represents an “anomaly”?



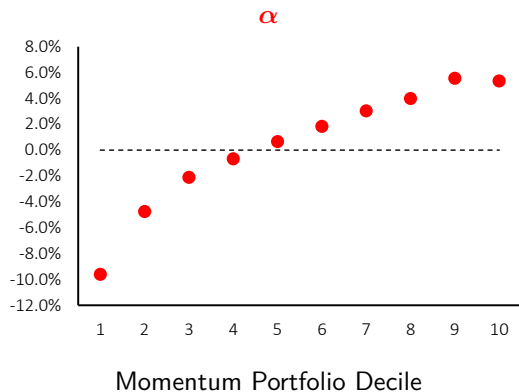
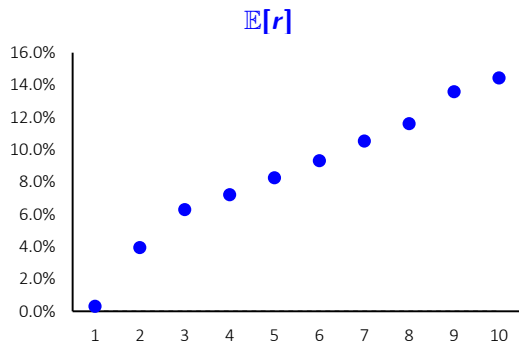
Source: Cochrane (2011)

- Several empirical decisions matter for the results:
 - How many portfolios?
 - * Terciles (3 portfolios) vs Quintiles (5 portfolios) vs Deciles (10 portfolios)
 - How to define the $s_{j,t}$ breakpoints?
 - * Using all stocks vs Using NYSE stocks
 - How to weight securities within a portfolio?
 - * Rank-Weights vs Equal-Weights vs Value-Weights
 - How often to rebalance the portfolios?
 - * Monthly vs Annual

Momentum Signals: Winners - Losers

- The principle underlying momentum signals:
 - Firms that outperformed in the recent past (winners) tend to provide higher future returns than firms that underperformed in the recent past (losers)
- Several momentum signals capture this logic. Examples:
 - Return from months $t - 12$ to $t - 1$
(Jegadeesh, Titman (2001))
 - Return from months $t - 12$ to $t - 7$
(Novy-Marx (2012))
 - Return from months $t - 6$ to $t - 1$ of the industry the firm belong to
(Moskowitz, Grinblatt (1999))
 - Abnormal return around recent past earnings announcements
(Chan, Jegadeesh, Lakonishok (1996))
 - Number of consecutive quarters with earnings increases
(Barth, Elliott, Finn (1999))
 - Recent past changes in analyst earnings forecasts
(Hawkins, Chamberlin, Daniel (1984))
 - Recent past earnings surprises
(Foster, Olsen, Shevlin (1984))
 - Recent past revenue surprises
(Jegadeesh, Livnat (2006))
 - Recent past returns of customers firms
(Cohen, Frazzini (2008))

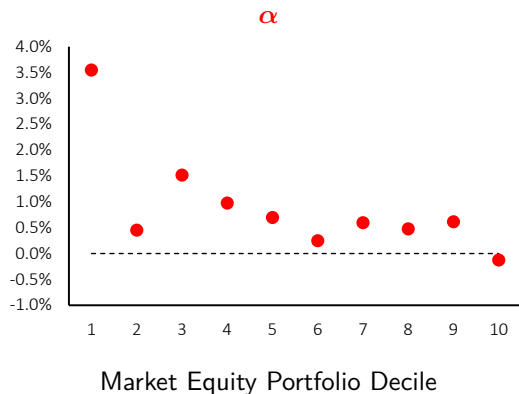
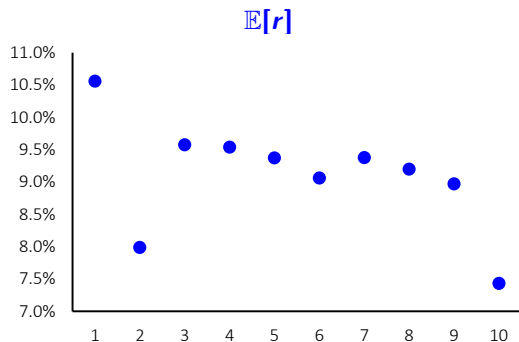
Momentum Signals: Winners - Losers



Friction Signals: High Frictions - Low Frictions

- The principle underlying friction signals:
 - Firms subject to a high level of frictions tend to provide higher future returns than firms subject to a low level of frictions
- Several friction signals capture this logic. Examples:
 - Size = Market Equity
(Banz (1981))
 - Price per share
(Miller, Scholes (1982))
 - Share turnover
(Datar, Naik, Radcliffe (1998))
 - Trading liquidity
(Amihud (2002))
 - Number of days with zero trade over recent months
(Liu (2006))
 - Bid-Ask spread implied from daily highest and lowest prices
(Corwin, Schultz (2012))
 - Dollar trading volume
(Brennan, Chordia, Subrahmanyam (1998))
 - Trading volume variance
(Chordia, Subrahmanyam, Anshuman (2001))
 - Exposure to systematic liquidity shocks
(Pastor, Stambaugh (2003))

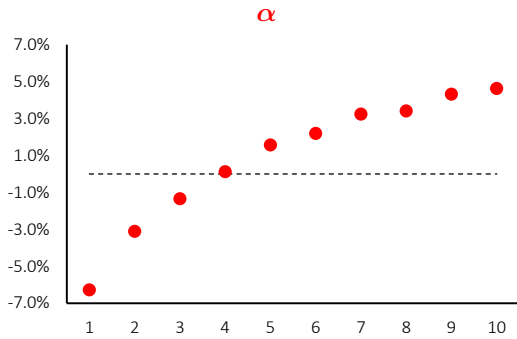
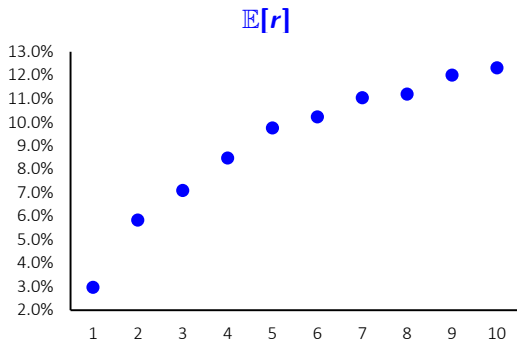
Friction Signals: High Frictions - Low Frictions



Value Signals: Value - Growth

- The principle underlying value signals:
 - Firms with low price relative fundamentals (value firms) tend to provide higher future returns than firms with high price relative to fundamentals (growth firms)
- Several value signals capture this logic. Examples:
 - Book Equity / Market Equity
(Rosenberg, Reid, Lanstein (1985), Fama, French (1992), Asness, Frazzini (2013))
 - Intangibles-Adjusted Book Equity / Market Equity
(Eisfeldt, Kim, Papanikolaou (2022), Park (2022))
 - Earnings / Market Equity
(Basu (1983), Elgers, Lo, Pfeiffer (2001))
 - Retained Earnings / Market Equity
(Ball et al (2020))
 - Equity Payout / Market Equity
(Boudoukh et al (2007))
 - Market Equity Projected on Fundamentals / Market Equity
(Bartram, Grinblatt (2018), Golubov, Konstantinidi (2019))
 - Fundamental Equity / Market Equity
(Frankel, Lee (1998), Gonçalves, Leonard (2023))
 - Normalized(Quality) - Normalized(Market-to-Book)
(Asness, Frazzini (2019))
 - Return from months $t - 60$ to $t - 13$ (return reversal reflects value)
(De Bondt, Thaler (1985))

Value Signals: Value - Growth

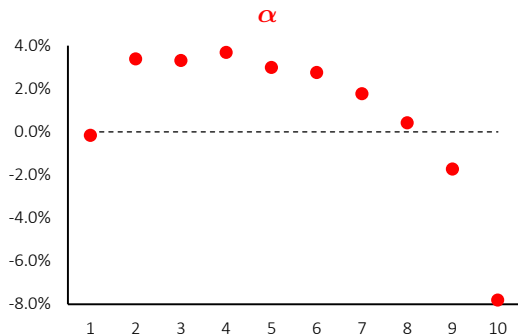
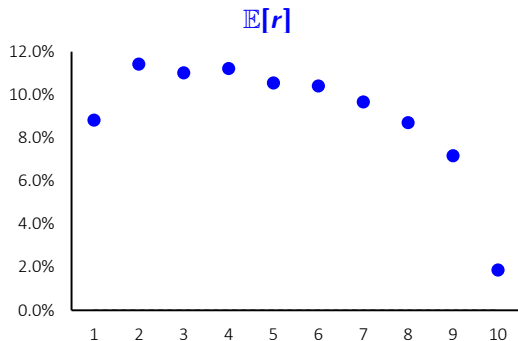


Book-to-Market Portfolio Decile

Investment Signals: Low Investment - High Investment

- The principle underlying investment signals:
 - Firms with low current investment tend to provide higher future returns than firms with high current investment
- Several investment signals capture this logic. Examples:
 - Asset Growth
(Cooper, Gulen, Schill (2008))
 - Change in PP&E and Inventory / Assets
(Lyandres, Sun, Zhang (2008))
 - Abnormal corporate investment
(Titman, Wei, Xie (2004))
 - Percentage change in investment relative to industry
(Abarbanell, Bushee (1998))
 - Inventory Investment
(Jones, Tuzel (2013))
 - Inventory Growth
(Belo, Lin (2012))
 - Labor hiring rate
(Belo, Lin, Bazdresch (2014))
 - Net Stock Issues
(Pontiff, Woodgate (2004))
 - Composite equity issuance
(Daniel, Titman (2006))

Investment Signals: Low Investment - High Investment

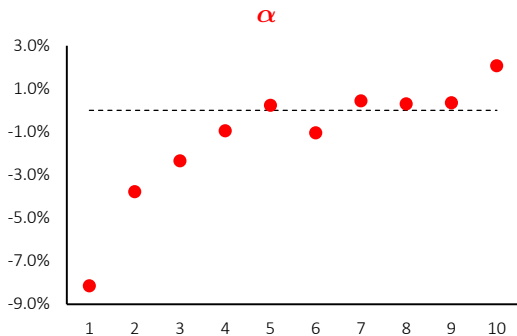
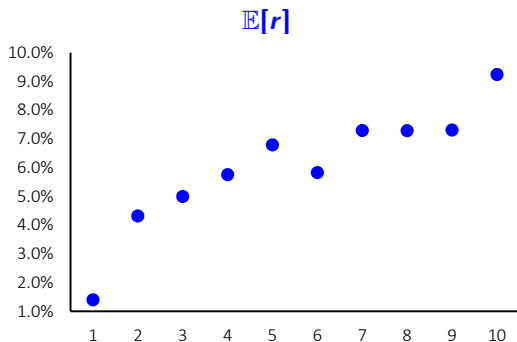


Asset Growth Portfolio Decile

Profitability Signals: High Profitability - Low Profitability

- The principle underlying profitability signals:
 - Firms with high profitability tend to provide higher future returns than firms with low profitability
- Several profitability signals capture this logic. Examples:
 - Return on Equity
(Haugen, Baker (1996))
 - Quarterly Return on Equity
(Hou, Xue, Zhang (2015))
 - Quarterly Return on Assets
(Balakrishnan, Bartov, Faurel (2010))
 - Return on Net Operating Assets
(Soliman (2008))
 - Gross Profitability
(Novy-Marx (2013))
 - Operating Profitability
(Fama, French (2015))
 - Cash-Based Profitability
(Ball et al (2016))
 - Profit Margin
(Soliman (2008))
 - Asset Turnover
(Soliman (2008))

Profitability Signals: High Profitability - Low Profitability



Quarterly Return on Equity Portfolio Decile

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Internal Validity: Publication Incentives

- “most claimed research findings in financial economics are likely false”

(Harvey, Liu, Zhu (2016))

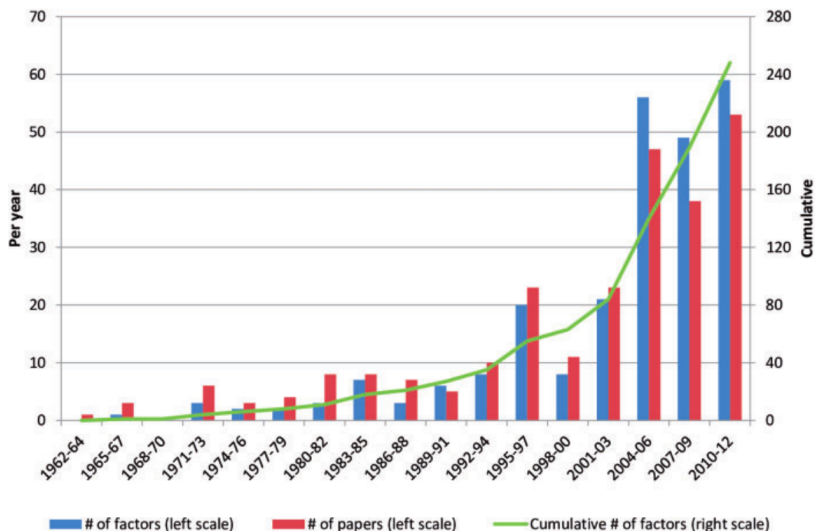


Figure 2
Factors and publications.

Source: Harvey, Liu, Zhu (2016)

Internal Validity: Publication Incentives

- Publication incentives are unlikely to explain all anomaly signals (Chen (2021))
 - With each researcher testing 8 signals per day, it takes 10,000 researchers hundreds of years to produce the published data
 - Moreover, researchers must have an extreme preference for very high t -stats

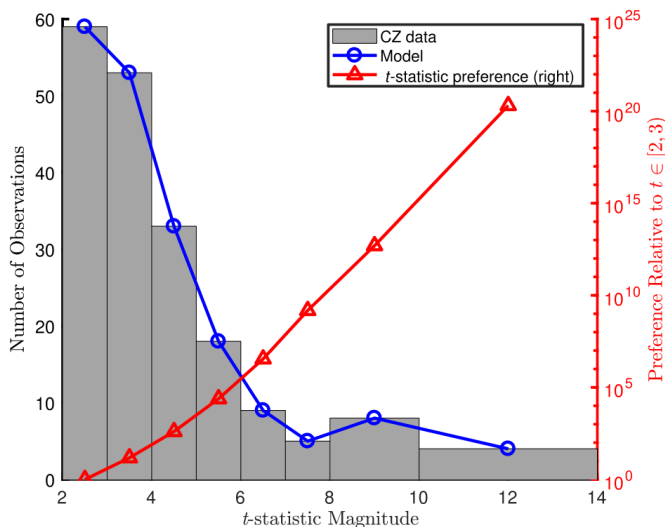


Figure 2. An extreme preference for large t -statistics

Internal Validity: Scientific Replication

- “most anomalies fail to replicate [using typical] standards for empirical finance”
(Hou, Xue, Zhang (2020))

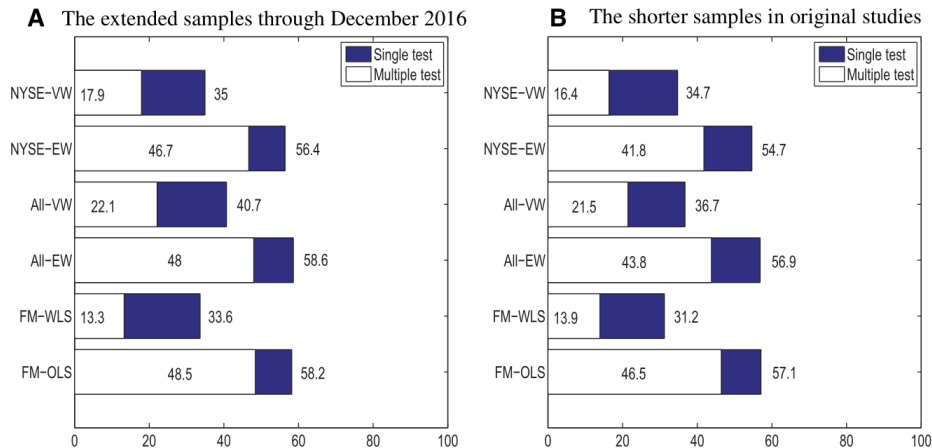


Figure 2
Replication rates (as a percentage), single and multiple tests, January 1967–December 2016, 600 months

Source: Hou, Xue, Zhang (2020)

Internal Validity: Scientific Replication

- “nearly 100% of the literature’s predictability results can be reproduced”

(Chen, Zimmermann (2022))

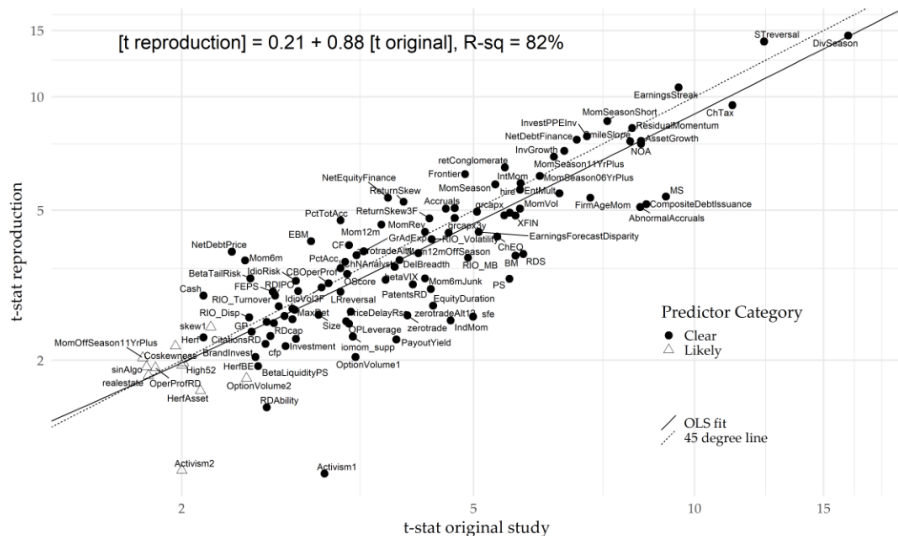


Figure 3: Comparison of Reproduced Long-Short t-Stats with Original Long-Short t-Stats for All Predictors.

Source: Chen, Zimmermann (2022)

Internal Validity: Scientific Replication

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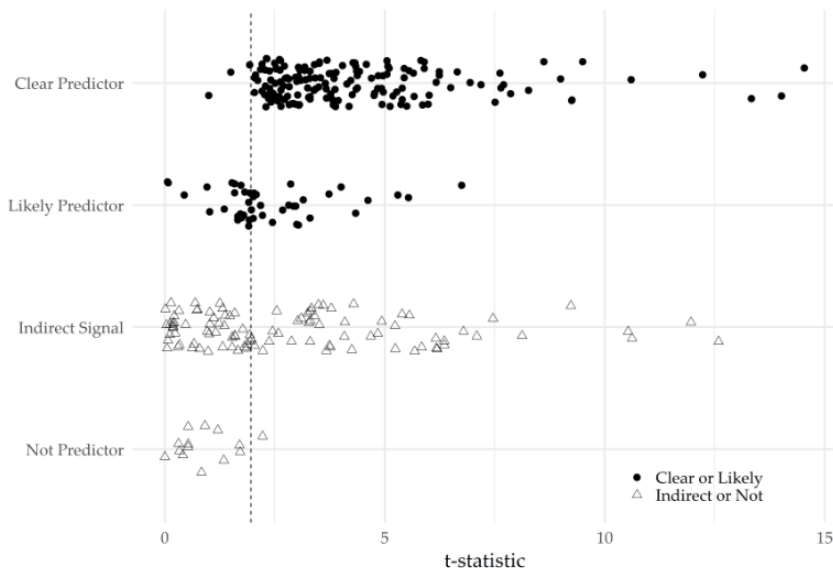
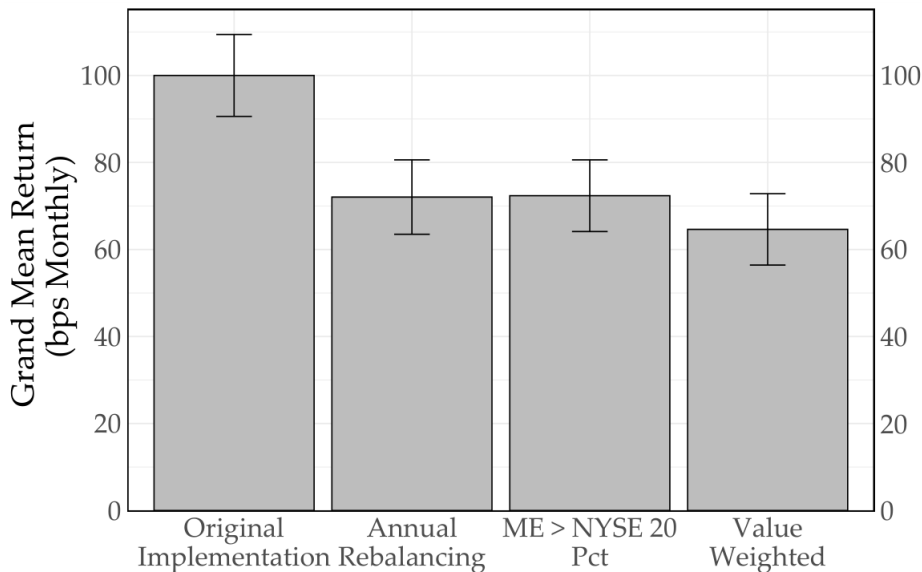


Figure 6: Reproduction Performance for All Characteristics

Source: Chen, Zimmermann (2022)

Internal Validity: Scientific Replication

- “nearly 100% of the literature's predictability results can be reproduced”
(Chen, Zimmermann (2022))



Source: Chen, Zimmermann (2023)

Anomaly Signals Pass Internal+External Validity Tests

- Anomaly signals pass internal+external validity tests (Jensen, Kelly, Pedersen (2023))
 - Scientific Replication: use a common set of empirical decisions
 - * Tercile breakpoints based on NYSE stocks with no microcaps
 - * Value-weights with market equity winsorized at 80th percentile
 - * One month holding period
 - Multiple Testing: use Bayesian method to jointly test anomaly signals
 - External Validity: extend sample to 93 countries

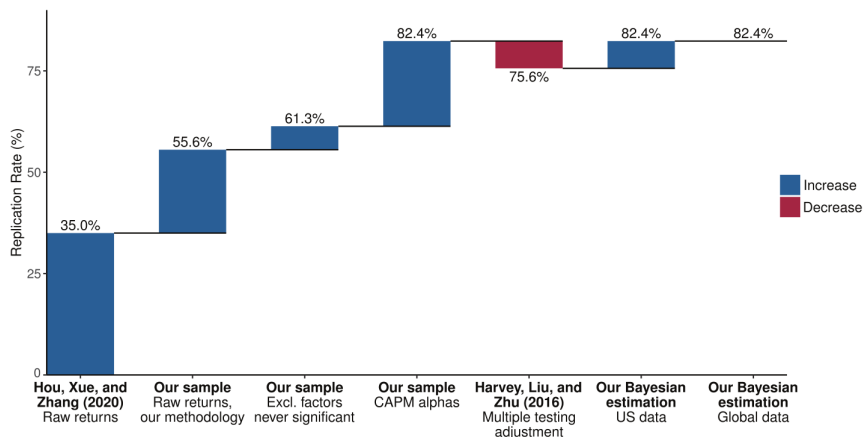
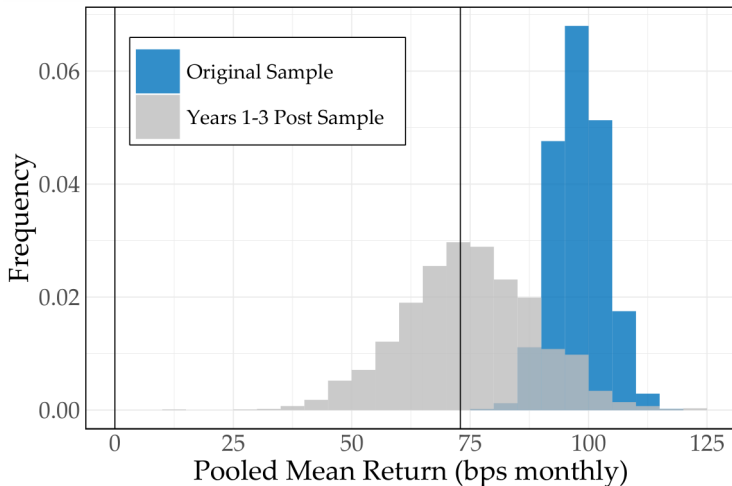


Figure 1. Replication rates versus the literature

Source: Jensen, Kelly, Pedersen (2023)

$\mathbb{E}[R]$ Decline Post Publication

- Anomaly signal returns tend to decline post publication (McClean, Pontiff (2016))

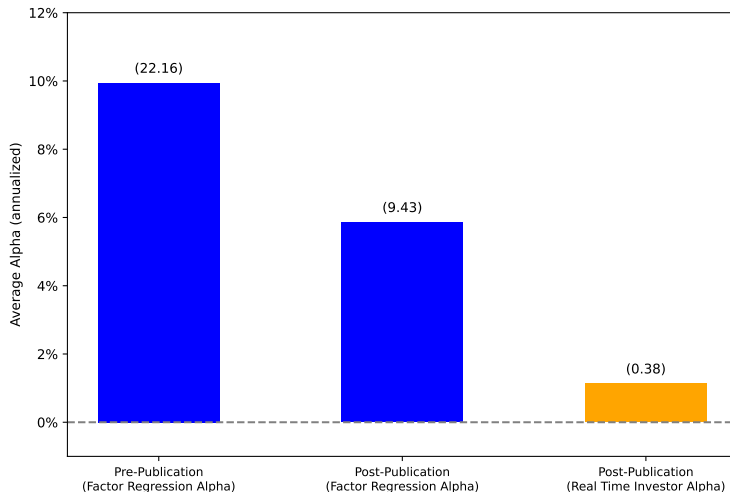


Source: Chen, Zimmermann (2023)

- Explanation (1): investors trade anomaly returns away post publication
 - US is the only country with reliable post publication decline (Jacobs, Müller (2020))
- Explanation (2): data mining incentives inflate published anomaly returns
 - Many anomalies disappear before publication sample (Linnainmaa, Roberts (2018))

Real Time Investor Alphas are Very Low (Extra)

- Alpha is connected to portfolio efficiency (in a Sharpe ratio sense)
(Gibbons, Ross, Shanken (1989))
- “Earning” the anomaly alpha requires building an optimal Sharpe ratio portfolio
(Cederburg, O’Doherty, Wang, Yan (2020))
- For a real time investor, the average post-publication alpha is very low
(Gonçalves, Loudis, Ogden (2026))

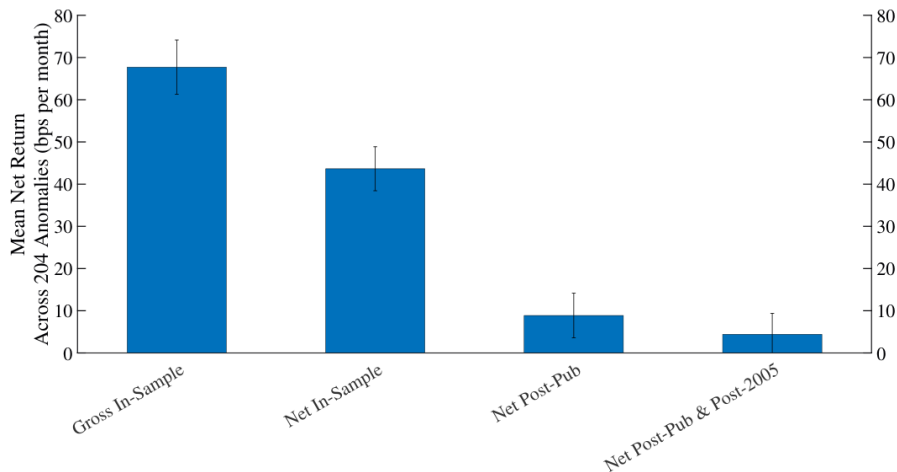


Source: Gonçalves, Loudis, Ogden (2026)

Trading Costs & Short-Selling Fees (Extra)

- The typical anomaly signal provides little $\mathbb{E}[R]$ after trading costs post 2005
(Chen, Velikov (2023))

Figure 1: Anomaly Mean Long-Short Returns. Error bars show two standard errors.



Source: Chen, Velikov (2023)

Trading Costs & Short-Selling Fees (Extra)

- The typical anomaly signal provides little $\mathbb{E}[R]$ after short-selling fees post 2005
(Muravyev, Pearson, Pollet (2023))

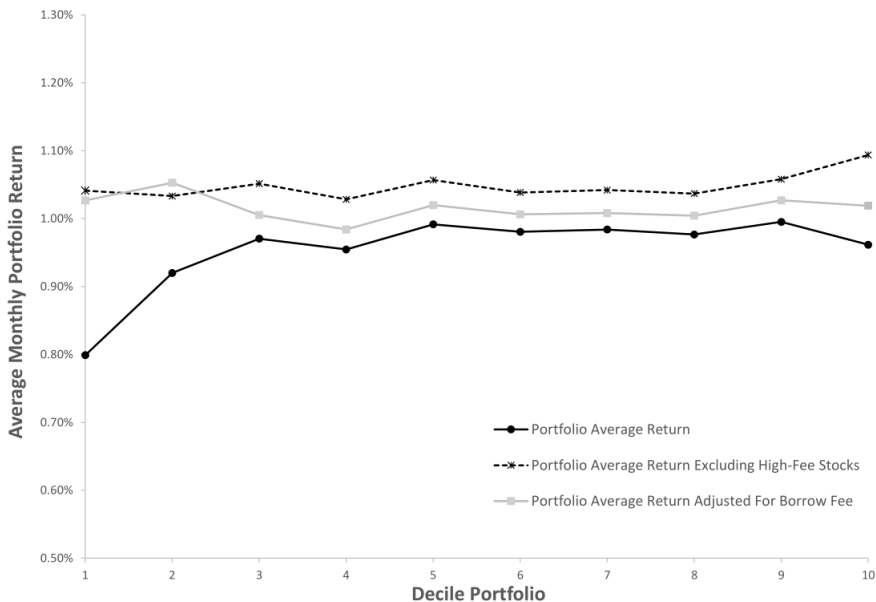


Figure 1. Average monthly return for each decile portfolio across anomalies

Muravyev, Pearson, Pollet (2023)

Anomaly Long-Term $\mathbb{E}[R]$ (Extra)

- Typical anomaly signal delivers no return after 5 years of portfolio formation
(Keloharju, Linnainmaa, Nyberg (2021))

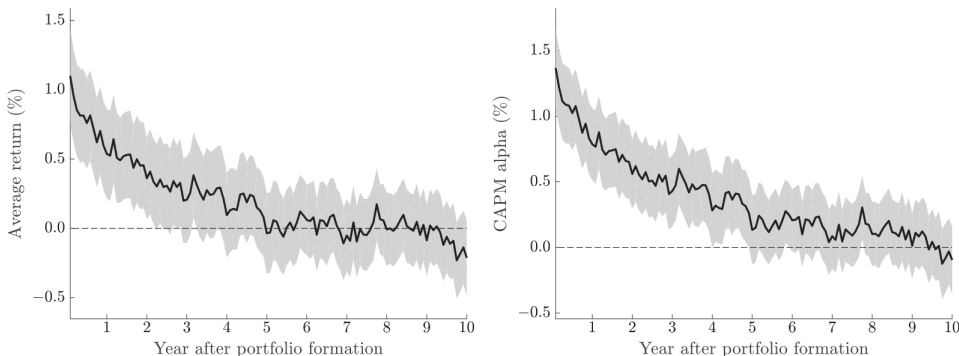


Fig. 5. Average monthly returns and CAPM alphas on long-short strategies in months 1–120 after portfolio formation.
Keloharju, Linnainmaa, Nyberg (2021)

- But we can construct signals that target (and deliver) long-run $\mathbb{E}[R]$
(Gonçalves, Leonard (2023), Gormsen, Huber, Jensen (2026))
- Cross-firm variation in long-run $\mathbb{E}[R]$ better reflects cross-firm price variation
(Cohen, Polk, Vuolteenaho (2009))
- Cross-firm variation in long-run $\mathbb{E}[R]$ is more important for capital allocation
(Binsbergen, Opp (2019))

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

- [2.1] Equity Premium Predictors
- [2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- [3.1] Equity Anomaly Signals
- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

(4) Bond Prices & $\mathbb{E}[R]$

- We saw that variation in $\mathbb{E}[R]$ has a large effect on equity prices
- We saw signals capturing equity $\mathbb{E}[R]$ variation (time series & cross section)
- We now explore:
 - (i) The effect of variation in bond $\mathbb{E}[R]$ on bond price variation
 - (ii) Signals capturing variation in bond $\mathbb{E}[R]$

[4.1] Treasury Bonds

- Both $\mathbb{E}[\text{Inflation}]$ and $\mathbb{E}[R]$ strongly affect bond prices
- Bond risk premia ($\mathbb{E}[xR]$) variation dominates time variation in bond $\mathbb{E}[R]$
- Signals from bond yields and macro variables capture bond $\mathbb{E}[xR]$ over time
- $\mathbb{E}[xR]$ vary across bonds based on cash flow duration and issuing country

[4.2] Corporate Bonds

- Both $\mathbb{E}[\text{Losses}]$ and $\mathbb{E}[xR]$ affect corporate bond prices
 - * $\mathbb{E}[xR]$ dominates for high rated bonds & $\mathbb{E}[\text{Losses}]$ dominates for low rate bonds
- The corporate yield spread captures $\mathbb{E}[xR]$ over time (especially long-run)
- Signals based on the credit and macroeconomic environment also do
- $\mathbb{E}[xR]$ vary across corporate bonds based on duration, rating, and liquidity
- Value signals as well as other signals also capture cross-bond variation in $\mathbb{E}[xR]$

Outline

(1) Variation in Equity Prices

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- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

Treasury Bond Strip Present Value Identity

- An H -year Treasury bond strip (i.e., zero-coupon bond)
 - Pays \$1 in H years (not inflation adjusted)
 - Has nominal price $\$P_{b,t}^{(H)} = \$1 \cdot e^{-H \cdot y_{b,t}^{(H)}}$
 - $y_{b,t}^{(H)} = -\frac{1}{H} \cdot \frac{\$P_{b,t}^{(H)}}{\$P_{b,t}^{(H)}}$ is the bond yield
 - $f_{b,t}^{(H)} = H \cdot y_{b,t}^{(H)} - (H-1) \cdot y_{b,t}^{(H-1)}$ is the bond forward rate
 - $\$R_{b,t+1}^{(H)} = \$P_{b,t+1}^{(H-1)} / \$P_{b,t}^{(H)}$ is the bond nominal return (note that $\$P_t^{(0)} = 1$)
 - Bond nominal log returns are linked to bond yields

$$\$r_{b,t+1}^{(H)} = \$P_{b,t+1}^{(H-1)} - \$P_{b,t}^{(H)} = H \cdot y_{b,t}^{(H)} - (H-1) \cdot y_{b,t+1}^{(H-1)}$$

- Bond real log returns adjust for inflation $r_{b,t+1}^{(H)} = \$r_{b,t+1}^{(H)} - \Delta\pi_{t+1}$
- Recursively substituting $\$p_b$ in the $\$r_b$ equation (Campbell, Ammer (1993)) ▶ (Proof)

$$\begin{aligned} y_{b,t}^{(H)} &= \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\$r_{b,t+h}^{(H-h+1)}] = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}] \\ &= \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}] \end{aligned}$$

- In shocks (with $N_{x,t}^{(H)} = \sum_{h=1}^H \Delta\mathbb{E}_t[x_{t+h}]$ reflecting news to x):

$$\widetilde{y}_{b,t}^{(H)} = (1/H) \cdot N_{xrb,t}^{(H)} + (1/H) \cdot N_{rb1,t}^{(H)} + (1/H) \cdot N_{\Delta\pi,t}^{(H)}$$

- Also note that $r_{b,t}^{(H)} = H \cdot y_{b,t}^{(H)} - (H-1) \cdot y_{b,t}^{(H-1)} - \Delta\pi_t$, and thus

$$\widetilde{r}_{b,t}^{(H)} = -(H-1) \cdot \widetilde{y}_{b,t}^{(H-1)} - \widetilde{\Delta\pi}_t \Rightarrow \widetilde{r}_{b,t}^{(H)} = -\widetilde{\Delta\pi}_t - N_{\Delta\pi,t}^{(H-1)} - N_{rb1,t}^{(H-1)} - N_{xrb,t}^{(H-1)}$$

Variation in Treasury Bond Prices

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

- When inflation is high, the real value of the bond cash flow decreases
- So, an increase in $\mathbb{E}[\text{Inflation}]$ works as a decline in $\mathbb{E}[\text{Cash Flow Growth}]$
 - $\uparrow$ **bond risk premia** $\Rightarrow \uparrow$ **bond yields**
 - $\uparrow$ **expected interest rate** $\Rightarrow \uparrow$ **bond yields**
 - $\uparrow$ **expected inflation** $\Rightarrow \uparrow$ **bond yields**
- Bond strips do not trade in the market so we recover them from coupon bonds
(see Gürkaynak, Sack, Wright (2007) and Liu, Wu (2021) for different methods)

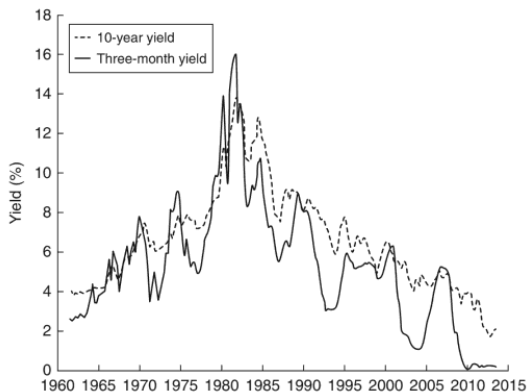


Figure 8.1. Three-Month and 10-Year Log Zero-Coupon US Treasury Yields, 1961–2013

Source: Campbell (2017), Chapter 8.1

Variation in Treasury Bond Prices

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

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 - $\uparrow$ **expected interest rate** $\Rightarrow$ $\uparrow$ **bond yields**
 - $\uparrow$ **expected inflation** $\Rightarrow$ $\uparrow$ **bond yields**
- Bond strips do not trade in the market so we recover them from coupon bonds (see Gürkaynak, Sack, Wright (2007) and Liu, Wu (2021) for different methods)
- Focus on shocks since bond yields are too persistent (Campbell, Ammer (1993))

$$\tilde{y}_{b,t}^{(H)} = (1/H) \cdot N_{xrb,t}^{(H)} + (1/H) \cdot N_{rb1,t}^{(H)} + (1/H) \cdot N_{\Delta\pi,t}^{(H)}$$

Decomposing $\text{Var}[\tilde{y}_b^{(10)}]$	1952-1987	1952-1972	1973-1987
% $\text{Var}[\tilde{y}_b^{(10)}]$ from N_{xrb}	49%	30%	57%
% $\text{Var}[\tilde{y}_b^{(10)}]$ from N_{rb1}	4%	0%	6%
% $\text{Var}[\tilde{y}_b^{(10)}]$ from $N_{\Delta\pi}$	47%	70%	37%

Source: Campbell and Ammer (1993), implied from Table IV

- So, both expected inflation and discount rates matter in explaining $\text{Var}[y_b^{(10)}]$
- Moreover, news to bond risk premia ($N_{xrb}^{(H)}$) dominate the discount rate part

The Expectation Hypothesis

- Expectation Hypothesis (EH): $\mathbb{E}_t[r_b^{(H)}] = \mathbb{E}[xr_b^{(H)}] + \mathbb{E}_t[r_b^{(1)}]$, which implies

$$\begin{aligned}
 y_{b,t}^{(H)} - y_{b,t}^{(1)} &= \text{const} + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] - \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[-\Delta\pi_{t+h}] - y_{b,t}^{(1)} \\
 &= \text{const} + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\mathbb{E}_{t+h-1}[r_{b,t+h}^{(1)} + \Delta\pi_{t+h}]] - y_{b,t}^{(1)} \\
 &= \text{const} + \frac{1}{H} \cdot \sum_{h=0}^{H-1} \mathbb{E}_t[y_{b,t+h}^{(1)} - y_{b,t}^{(1)}] \\
 &= \text{const} + \frac{1}{H} \cdot \sum_{h=1}^{H-1} (H-h) \cdot \mathbb{E}_t[\Delta y_{b,t+h}^{(1)}]
 \end{aligned}$$

- EH: term spread ($y_{b,t}^{(H)} - y_{b,t}^{(1)}$) tells us the expected path of future interest rates
- Many ways to test the EH and typical tests reject it
 - 49% of $\text{Var}[\widetilde{y}_b^{(H)}]$ is due to $N_{xrb,t}^{(H)} \Rightarrow$ EH rejection (Campbell, Ammer (1993))
 - We can predict future interest rates (implication: $b_H = 1$) (Campbell, Shiller (1991))

$$\frac{1}{H} \cdot \sum_{h=1}^{H-1} (H-h) \cdot \Delta y_{b,t+h}^{(1)} = a_H + b_H \cdot (y_{b,t}^{(H)} - y_{b,t}^{(1)}) + \varepsilon_t^{(H)}$$

H (in years) =	2		3		4		5	
b_H [t_{stat}]	0.24	[3.46]	0.41	[2.27]	0.56	[1.87]	0.68	[1.54]
R^2	1.1%		3.4%		7.2%		11.4%	

Source: Kojien, Nieuwerburgh (2021), term structure of interest rates lecture notes

$\mathbb{E}[R]$ Time Series Variation: Bond Yields

- EH Failure $\Rightarrow$ Treasury bond $\mathbb{E}_t[xr_b^{(H)}]$ varies over time
- Bond yields (or forward rates = transformation of bond yields) capture $\mathbb{E}_t[xr_b^{(H)}]$
 - Regressions of long-term excess returns on yield spreads (Campbell, Shiller (1991))

$$(1/H) \cdot xr_{b,t \rightarrow t+H}^{(H)} = a_H + b_H \cdot (y_{b,t}^{(H)} - y_{b,t}^{(1)}) + \varepsilon_t^{(H)}$$

- Regressions of 1-year excess returns on forward spreads (Fama, Bliss (1987))

$$xr_{b,t+1}^{(H)} = a_H + b_H \cdot (f_{b,t}^{(H)} - f_{b,t}^{(1)}) + \varepsilon_t^{(H)}$$

H =		2 years	3 years	4 years	5 years
Yields	$b_H [t_{stat}]$	0.76 [3.46]	0.59 [2.27]	0.44 [1.87]	0.32 [1.54]
	R^2	9.5%	6.9%	4.6%	2.9%
Forwards	$b_H [t_{stat}]$	0.76 [3.46]	1.00 [3.59]	1.27 [4.03]	1.06 [3.00]
	R^2	9.5%	10.6%	13.6%	7.9%

Source: Kojien, Nieuwerburgh (2021), term structure of interest rates lecture notes

- We can create a predictor that combines forward rates (Cochrane, Piazzesi (2005))
 - Estimate a single regression based on forward rates

$$\frac{1}{4} \cdot \sum_{H=2}^5 xr_{b,t+1}^{(H)} = \theta_0 + \sum_{H=1}^5 \theta_H \cdot f_{b,t}^{(H)} + u_t$$

- Use $cp_t = \sum_{H=1}^5 \hat{\theta}_H \cdot f_{b,t}^{(H)}$ as a single bond predictor

$$xr_{b,t+1}^{(H)} = a_H + b_H \cdot cp_t + \varepsilon_t^{(H)}$$

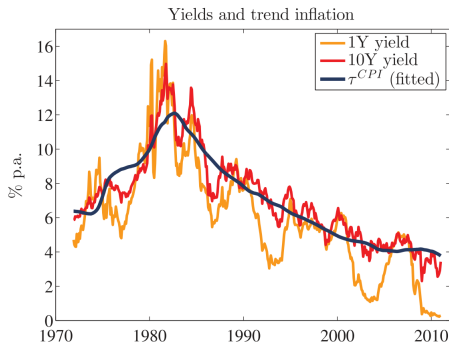
H =	2 years	3 years	4 years	5 years
b_H (SE)	0.47 (0.02)	0.87 (0.02)	1.24 (0.02)	1.43 (0.03)
R^2	31%	34%	37%	34%

Source: Cochrane, Piazzesi (2005), Table 1

$\mathbb{E}[R]$ Time Series Variation: Cyclical Parts of Bond Yields

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

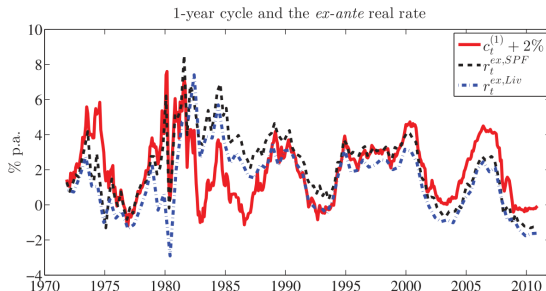
- Bond yields have a persistent component related to **inflation**
- Cyclical parts of bond yields relate to the **real interest rate** and **bond risk premia**
- We can separate the three components of $y_{b,t}^{(H)}$ (Cieslak, Povala (2015))
 - $\tau_t = (1 - \nu) \cdot \sum_{h=0}^{h=t-1} \nu^h \cdot \Delta\pi_{t-h}$ (exponential weighted moving average)
 - $y_{b,t}^{(H)} = \theta_{0,H} + \theta_H \cdot \tau_t + \eta_t^{(H)}$ (regression of bond yields on τ_t)
 - Bond yield cycle is $c_t^{(H)} = \hat{\eta}_t^{(H)}$ (estimated residual)
 - Define $\bar{c}_t = \frac{1}{19} \cdot \sum_{H=2}^{20} c_t^{(H)}$ (average bond yield cycle)
- τ_t captures the **inflation** component



$\mathbb{E}[R]$ Time Series Variation: Cyclical Parts of Bond Yields

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

- Bond yields have a persistent component related to **inflation**
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 - Bond yield cycle is $c_t^{(H)} = \hat{\eta}_t^{(H)}$ (estimated residual)
 - Define $\bar{c}_t = \frac{1}{19} \cdot \sum_{H=2}^{20} c_t^{(H)}$ (average bond yield cycle)
- τ_t captures the **inflation** component
- $c_t^{(1)}$ captures the **real interest rate** component



$\mathbb{E}[R]$ Time Series Variation: Cyclical Parts of Bond Yields

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[xr_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

- Bond yields have a persistent component related to **inflation**
- Cyclical parts of bond yields relate to the **real interest rate** and **bond risk premia**
- We can separate the three components of $y_{b,t}^{(H)}$ (Cieslak, Povala (2015))
 - $\tau_t = (1 - \nu) \cdot \sum_{h=0}^{t-1} \nu^h \cdot \Delta\pi_{t-h}$ (exponential weighted moving average)
 - $y_{b,t}^{(H)} = \theta_{0,H} + \theta_H \cdot \tau_t + \eta_t^{(H)}$ (regression of bond yields on τ_t)
 - Bond yield cycle is $c_t^{(H)} = \hat{\eta}_t^{(H)}$ (estimated residual)
 - Define $\bar{c}_t = \frac{1}{19} \cdot \sum_{H=2}^{20} c_t^{(H)}$ (average bond yield cycle)
- τ_t captures the **inflation** component
- $c_t^{(1)}$ captures the **real interest rate** component
- $\bar{c}_t$ captures the **bond risk premia** component
- To explore this **bond risk premia** component, consider the regressions of the form

$$\frac{1}{19} \cdot \sum_{H=2}^{20} \frac{1}{H} \cdot xr_{b,t+1}^{(H)} = a + b' s_t + u_t$$

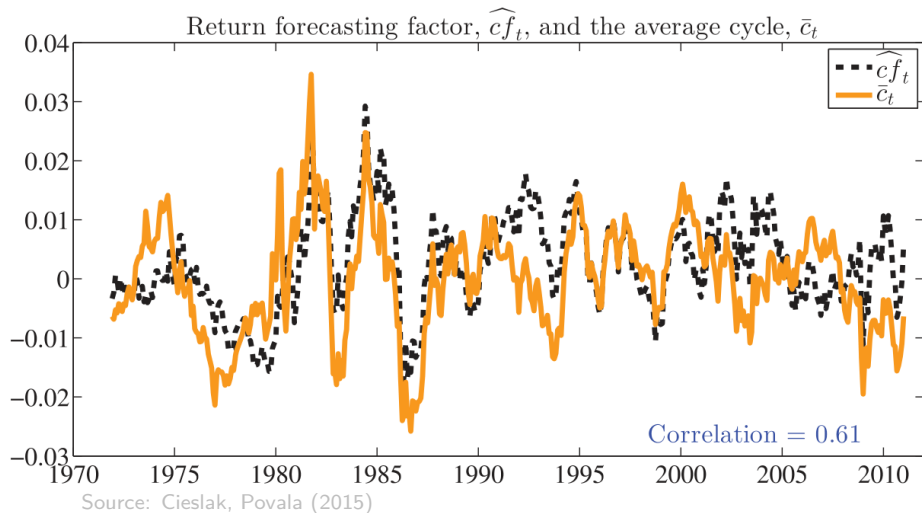
$\mathbb{E}[R]$ Time Series Variation: Cyclical Parts of Bond Yields

Table 2 Predictive regressions

Regressors $\rightarrow$	Yields only (1)	Yields+ τ^{CPI} (2)	$\tilde{y}_t, y_t^{(1)}$ (3)	$\tilde{y}_t, y_t^{(1)}, \tau_t^{CPI}$ (4)	$\tilde{c}_t, c_t^{(1)}$ (5)
<i>Regression coefficients</i>					
$y^{(1)}$ or $c^{(1)}$	−1.13 (−1.87)	−1.09 (−1.64)	−0.42 (−2.48)	−0.61 (−3.70)	−0.61 (−3.67)
$y^{(2)}$ or $c^{(2)}$	0.73 (0.62)	1.06 (0.81)	— —	— —	— —
$y^{(5)}$ or $c^{(5)}$	0.83 (0.99)	−0.71 (−0.10)	— —	— —	— —
$y^{(7)}$ or $c^{(7)}$	0.40 (0.15)	0.51 (0.32)	— —	— —	— —
$y^{(10)}$ or $c^{(10)}$	−1.15 (−1.69)	0.84 (0.43)	— —	— —	— —
$y^{(20)}$ or $c^{(20)}$	0.37 (0.94)	0.21 (0.49)	— —	— —	— —
τ^{CPI}	— —	−1.02 (−4.30)	— —	−1.01 (−4.65)	— —
$\bar{y}$ or $\bar{c}$	— —	— —	0.54 (2.47)	1.45 (5.03)	1.45 (5.03)
$\bar{R}^2$	0.24	0.54	0.18	0.53	0.53

Source: Cieslak, Povala (2015)

$\mathbb{E}[R]$ Time Series Variation: Cyclical Parts of Bond Yields



$\mathbb{E}[R]$ Time Series Variation: Macroeconomic Variables

- Macroeconomic variables add to bond yields in capturing $\mathbb{E}_t[xr_b^{(H)}]$
- Challenge: too many macroeconomic variables
- Diffusion Index Forecasting with large macro dataset (Ludvigson, Ng (2009))
 - Large dataset with 132 measures of economic activity
(close to what is now the MACRO-MD dataset, see McCracken, Ng (2016))
 - Estimate 8 principal components ($\widehat{F}_t$) using a selection criterium (Bai, Ng (2002))
 - Predict $xr_{b,t+1}^{(H)}$ using subset of $\widehat{F}_t$ (selected by BIC) controlling for cp_t
 - Linear combination of $\widehat{F}_t$ subset (called $F5_t$) is a good summary of $\widehat{F}_t$

Table 2 Regression of monthly excess bond returns on lagged factors

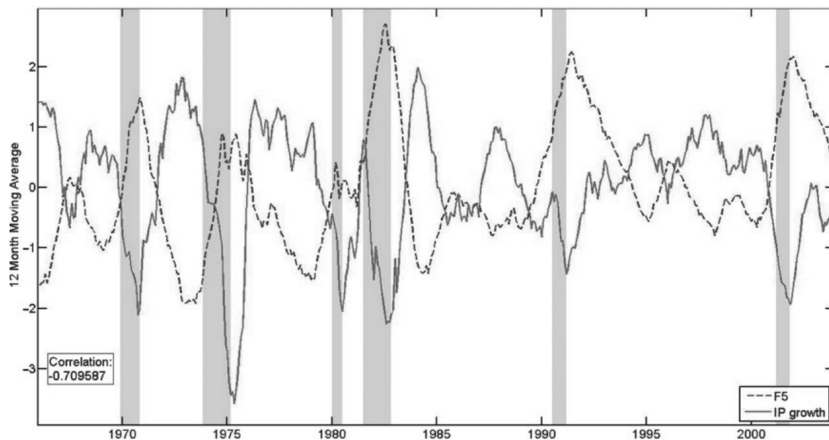
$$\text{Model: } rx_{t+1}^{(n)} = \beta_0 + \beta_1' \widehat{F}_t + \beta_2 CP_t + \epsilon_{t+1}$$

	CP_t	$F5_t$	$\bar{R}^2$		CP_t	$F5_t$	$\bar{R}^2$
$rx_{t+1}^{(2)}$	0.45 (8.90)		0.31	$rx_{t+1}^{(3)}$	0.85 (8.52)		0.34
		0.54 (5.52)	0.22			0.91 (5.28)	0.19
	0.39 (6.0)	0.43 (5.78)	0.44		0.75 (6.16)	0.69 (5.55)	0.44
$rx_{t+1}^{(4)}$	1.24 (8.58)		0.37	$rx_{t+1}^{(5)}$	1.46 (7.90)		0.34
		1.19 (5.08)	0.17			1.36 (4.80)	0.14
	1.11 (6.30)	0.87 (5.39)	0.45		1.32 (5.87)	0.98 (5.08)	0.42

Source: Ludvigson, Ng (2009)

$\mathbb{E}[R]$ Time Series Variation: Macroeconomic Variables

- Macroeconomic variables add to bond yields in capturing $\mathbb{E}_t[xr_b^{(H)}]$
- Challenge: too many macroeconomic variables
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 - Linear combination of $\hat{F}_t$ subset (called $F5_t$) is a good summary of $\hat{F}_t$



Source: Ludvigson, Ng (2009)

$\mathbb{E}[R]$ Time Series Variation: Macroeconomic Variables

- Macroeconomic variables add to bond yields in capturing $\mathbb{E}_t[xr_b^{(H)}]$
- Challenge: too many macroeconomic variables
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 - Predict $xr_{b,t+1}^{(H)}$ using subset of $\widehat{F}_t$ (selected by BIC) controlling for cp_t
 - Linear combination of $\widehat{F}_t$ subset (called $F5_t$) is a good summary of $\widehat{F}_t$
- Machine Learning with large macro dataset (Bianchi, Büchner, Tamoni (2021))
 - Treasury bond yield curve data with maturities up to 10 years (Liu, Wu (2021))
 - MACRO-MD dataset (McCracken, Ng (2016))
 - Use machine learning methods to forecast $xr_{b,t+1}^{(H)}$ (with and without macro)
 - Evaluate methods based on R_{OS}^2
 - Dimensionality reduction methods (PCA and PLS) perform poorly
 - Penalized regression methods (Ridge, Lasso, Elastic Net) perform ok
 - Non-linear methods (regression trees and neural networks) perform very well
 - Adding macro variables drastically improves R_{OS}^2
 - $\mathbb{E}_t[xr_b^{(H)}]$ is countercyclical (as in Ludvigson, Ng (2009), Cieslak, Povala (2015))

$\mathbb{E}[R]$ Cross Sectional Variation

- Treasury bonds vary with
 - Cash flow duration (i.e., weighted average of cash flow maturities)
 - Country issuing the bond
- Cash flow duration
 - Bonds with higher duration have higher $\mathbb{E}[R]$ (the “bond term premium”)
 - This pattern reflects an upward sloping term structure of interest rates

Table 8.1. Means and Standard Deviations of Term-Structure Variables, 1961–2013

Variable	Long bond maturity (n), quarterly intervals					
	2	3	4	8	20	40
Excess return $r_{n,t+1} - y_{1t}$	0.117 (0.878)	0.283 (1.591)	0.426 (2.284)	0.865 (4.696)	1.650 (10.030)	2.173 (17.052)
Yield spread $y_{nt} - y_{1t}$	0.052 (0.253)	0.121 (0.368)	0.187 (0.458)	0.410 (0.713)	0.846 (1.119)	1.259 (1.456)

Source: Campbell (2017), Chapter 8.1

- Country issuing the bond
 - Countries w/ lower credit rating (higher default probability) have higher bond $\mathbb{E}[R]$
 - Countries with higher bond value and/or bond momentum have higher bond $\mathbb{E}[R]$
(bond value = 5-year changes in $y_t^{(10)}$; bond momentum = bond index $R_{t-12 \rightarrow t-1}$)

	Value Portfolios				Momentum Portfolios				50/50 Combination
	P1	P2	P3	P3-P1	P1	P2	P3	P3-P1	P3-P1
Mean	3.0%	4.0%	4.2%	1.1%	3.8%	3.8%	4.2%	0.4%	0.8%
(t -stat)	(2.31)	(3.58)	(3.76)	(0.97)	(3.42)	(3.49)	(3.28)	(0.35)	(1.03)
Stdev	7.0%	5.9%	5.9%	6.3%	5.9%	5.9%	6.8%	6.0%	4.0%
Sharpe	0.43	0.67	0.71	0.18	0.64	0.66	0.61	0.06	0.19

Source: Asness, Moskowitz, Pedersen (2013)

Outline

(1) Variation in Equity Prices

- [1.1] The Present Value Identity
- [1.2] Variation in Aggregate Equity Prices
- [1.3] Variation in Firm-Level Equity Prices
- [1.4] Important Issues and Developments

(2) Equity $\mathbb{E}[R]$ (Time Series)

- [2.1] Equity Premium Predictors
- [2.2] Important Issues and Developments

(3) Equity $\mathbb{E}[R]$ (Cross Section)

- [3.1] Equity Anomaly Signals
- [3.2] Important Issues and Developments

(4) Bond Prices & $\mathbb{E}[R]$

- [4.1] Treasury Bonds
- [4.2] Corporate Bonds

Variation in Corporate Bond Prices

- Consider a coupon-paying Corporate bond (cb) with H -year maturity at t
- Consider a Treasury bond (tb) with the same structure (maturity and cash flows)
- We can approximate the log return spread between the two as (Nozawa (2017))

$$\log(R_{cb,t+1}) - \log(R_{tb,t+1}) = \textcolor{red}{x}r_{cb,t+1} \approx \text{const} - \kappa \cdot \textcolor{green}{p}s_{cb,t+1} + \textcolor{green}{p}s_{cb,t} - \textcolor{blue}{l}_{cb,t+1}$$

where

$\textcolor{green}{p}s_{cb,t} = \log(P_{tb,t}/P_{cb,t})$ if not defaulted or matured (0 otherwise)

$\textcolor{blue}{l}_{cb,t} = \log(P_{tb,\tau}/P_{cb,\tau})$ at default time $t = \tau$ (0 otherwise)

- Recursively substituting $\textcolor{green}{p}s_{cb,t+h}$, we have (Nozawa (2017))

► (Proof)

$$\textcolor{green}{p}s_{cb,t} \approx \text{const} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\textcolor{red}{x}r_{cb,t+h}] + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\textcolor{blue}{l}_{cb,t+h}]$$

Decomposing Firm Level $\text{Var}[\textcolor{green}{p}s_{cb}]$	A+	Baa	Ba	B-
% $\text{Var}[\textcolor{green}{p}s_{cb}]$ from $\mathbb{E}[\textcolor{red}{x}r_{cb}]$	90%	81%	54%	12%
% $\text{Var}[\textcolor{green}{p}s_{cb}]$ from $\mathbb{E}[\textcolor{blue}{l}_{cb}]$	9%	18%	46%	89%
Aggregate Volatility / Firm Volatility	$\mathbb{E}[\textcolor{red}{x}r_{cb}]$ Component 1.17		$\mathbb{E}[\textcolor{blue}{l}_{cb}]$ Component 0.08	

Source: Nozawa (2017), implied from Tables II and IV

Variation in Corporate Bond Prices

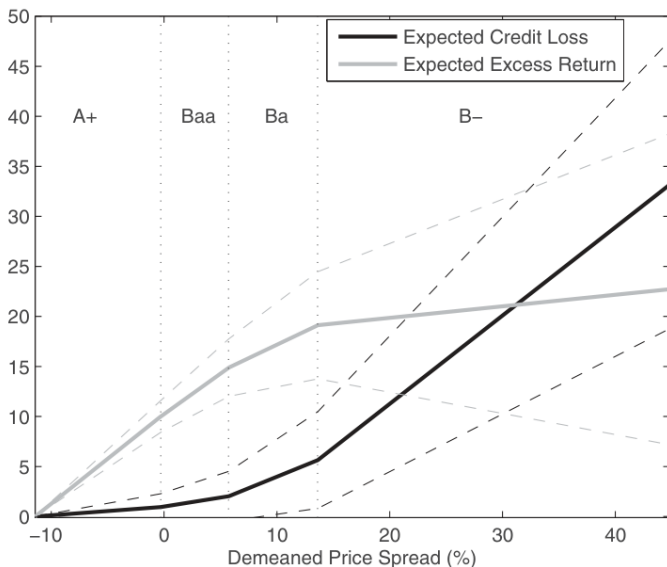
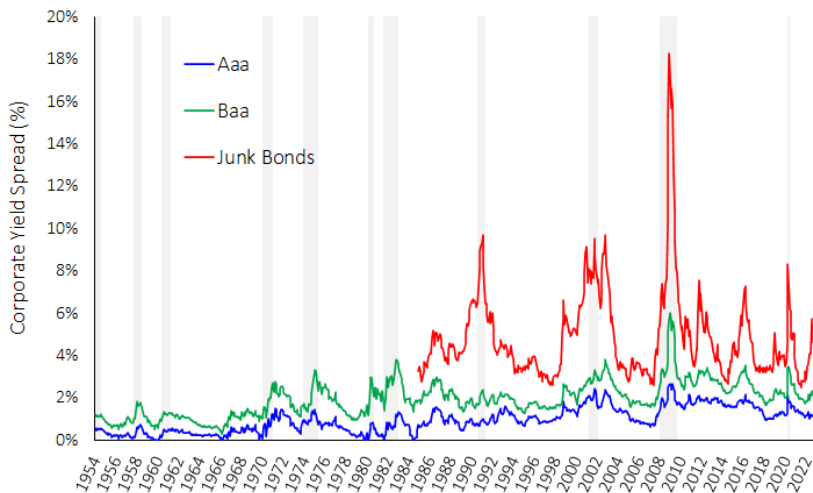


Figure 1. Long-run forecasting coefficients by credit ratings. Along the x-axis is demeaned credit spreads, with the left end set by the 1st percentile of credit spreads and the right end set by the 99th percentile. Along the y-axis is the long-run expected credit loss, $E_t[\sum \rho^{j-1} l_{i,t+j}]$, and excess return, $E_t[\sum \rho^{j-1} \bar{r}_{i,t+j}]$, with the slope given by the long-run forecasting coefficients in Panel A of Table II. Dashed lines denote $\pm$ standard error bounds. The borders between credit ratings are located where the histogram of the credit spread for a given credit rating overlaps with the histogram of the credit spread for the neighboring credit ratings.

Source: Nozawa (2017)

$\mathbb{E}[R]$ Time Series Variation

- Aggregate corporate bond expected returns vary over time
- Aggregate corporate yield spread ($y_{cb} - y_{tb}$) (Keim, Stambaugh (1986))
 - It is a strong predictor of long-term bond returns
 - The present value identity of the prior slide provides the motivation
 - The evidence in the prior slide quantifies the importance of this predictability

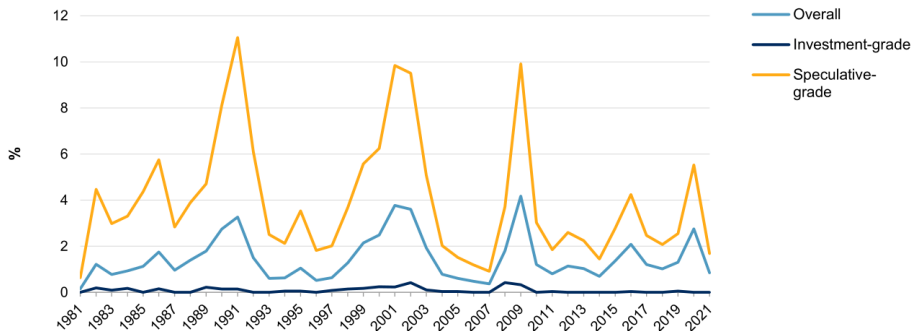


Source: My calculation w/ data from the Federal Reserve Economic Data (FRED)

$E[R]$ Time Series Variation

- Aggregate corporate bond expected returns vary over time
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Global Default Rates: Investment-Grade Versus Speculative-Grade



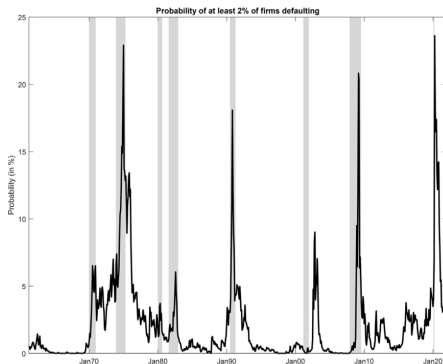
Sources: S&P Global Ratings Research and S&P Global Market Intelligence's CreditPro®.

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Source: S&P Global 2021 Annual Global Corporate Default and Rating Transition Study

$\mathbb{E}[R]$ Time Series Variation

- Aggregate corporate bond expected returns vary over time
- Aggregate corporate yield spread ($y_{cb} - y_{tb}$) (Keim, Stambaugh (1986))
 - It is a strong predictor of long-term bond returns
 - The present value identity of the prior slide provides the motivation
 - The evidence in the prior slide quantifies the importance of this predictability
- There are several other aggregate corporate bond return predictors
 - Systematic default probability (Bao, Hou, Zhang (2023))
 - Aggregate corporate bond return volatility (Bai, Bali, Wen (2021))
 - Relative quantity of long-term bond issuances (Baker, Greenwood, Wurgler (2003))
 - Relative credit quality of aggregate bond issuances (Greenwood, Hanson (2013))
 - Combination of macro, equity, and bond variables (Lin, Wu, Zhou (2017))



Source: Bao, Hou, Zhang (2023)

$\mathbb{E}[R]$ Cross Sectional Variation: Duration+Rating+Liquidity

- Bonds largely vary with
 - Cash flow duration (i.e., weighted average of cash flow maturities)
 - Credit rating (i.e., perceived default probability)
- Not surprisingly, bond expected returns vary in these two dimensions

Table 5

Intermediate	AAA	AA	A	BAA
Average excess return	2.38%	2.53%	2.76%	3.44%
Long term	AAA	AA	A	BAA
Average excess return	3.12%	3.80%	3.75%	4.60%

Source: Binsbergen, Kojien (2017)

$\mathbb{E}[R]$ Cross Sectional Variation: Duration+Rating+Liquidity

- Bonds largely vary with
 - Cash flow duration (i.e., weighted average of cash flow maturities)
 - Credit rating (i.e., perceived default probability)
- Not surprisingly, bond expected returns vary in these two dimensions
- Bond expected returns also vary with liquidity (but harder to measure liquidity)
(Longstaff, Mithal, Neis (2005), Chen, Lesmond, Wei (2007), Lin, Wang, Wu (2011), Dick-Nielsen, Feldhütter, Lando (2012), Bongaerts, de Jong, Driessen (2017))

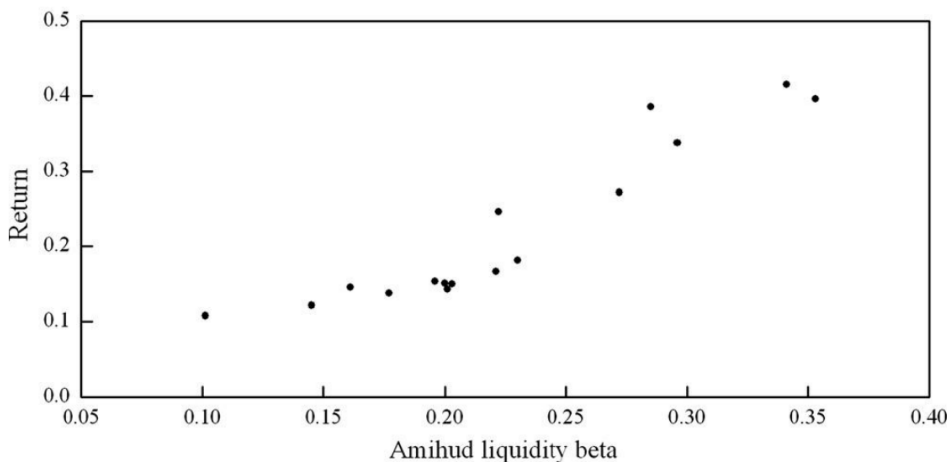


Fig. 2. The relation between bond excess returns and liquidity beta.

Source: Lin, Wang, Wu (2011)

$\mathbb{E}[R]$ Cross Sectional Variation: Duration+Rating+Liquidity

- Bonds largely vary with
 - Cash flow duration (i.e., weighted average of cash flow maturities)
 - Credit rating (i.e., perceived default probability)
- Not surprisingly, bond expected returns vary in these two dimensions
- Bond expected returns also vary with liquidity (but harder to measure liquidity)

(Longstaff, Mithal, Neis (2005), Chen, Lesmond, Wei (2007), Lin, Wang, Wu (2011), Dick-Nielsen, Feldhütter, Lando (2012), Bongaerts, de Jong, Driessen (2017))
- The liquidity effect was particularly strong during the great recession

(Dick-Nielsen, Feldhütter, Lando (2012))

Table 3 Liquidity regressions.

For each rating class R and each liquidity variable L a pooled regression is run with credit risk controls

$$\text{Spread}_{it}^R = \alpha^R + \gamma^R L_{it} + \text{Credit risk controls}_{it} + \epsilon_{it},$$

Panel A: Pre-subprime (2005:Q1–2007:Q1)

	AAA	AA	A	BBB	Spec
λ	0.0038*** (2.97)	0.0056*** (2.95)	0.0131*** (2.61)	0.0260*** (3.69)	0.1726*** (5.34)

Panel B: Post-subprime (2007:Q2–2009:Q2)

	AAA	AA	A	BBB	Spec
λ	0.0281** (2.12)	0.2495*** (3.64)	0.2500*** (4.08)	0.3333*** (3.57)	0.6746*** (6.73)

Source: Dick-Nielsen, Feldhütter, Lando (2012)

$\mathbb{E}[R]$ Cross Sectional Variation: Value Signals

- As in equities, value signals capture long-term $\mathbb{E}[R]$ of corporate bonds
- But measuring value is more challenging in corporate bonds
- Several bond-level value signals have been used in the literature:
 - Percentage difference between actual and theoretical corporate yield spreads (Correia, Richardson, Tuna (2012))
 - Percentage difference between actual and fitted corporate yield spreads (fitted based on bond-level characteristics) (Houweling, Zundert (2017))
 - Standardized difference between actual and fitted corporate yield spreads (fitted based on aggregate equity and bond factors) (Guo et al (2022))
 - Past long-term bond returns (months $t - 48$ to $t - 13$) (Bali, Subrahmanyam, Wen (2021))

Table 2: PE quintile portfolios: Returns

Rating	Low	2	3	4	High	H-L	t -stats
All	0.32	0.60	0.68	0.84	1.24	0.92	8.99
AAA+AA	0.32	0.57	0.63	0.75	0.98	0.66	10.73
A	0.32	0.58	0.66	0.71	1.02	0.70	7.67
BBB	0.37	0.60	0.73	0.85	1.38	1.01	7.59
Junk	0.21	0.71	0.99	1.48	1.83	1.62	4.86

Source: Guo et al (2022)

$\mathbb{E}[R]$ Cross Sectional Variation: Other Signals (Extra)

- Several other signals capture cross sectional variation in corporate bond $\mathbb{E}[R]$
(see Section 2.2 of Huang, Shi (2021) for a literature review)
- They can be broadly divided into bond-level signals and issuer-level signals
- Bond-level signals
 - Bond past mid-term returns (bond momentum)
(Pospisil, Zhang (2010), Jostova et al (2013))
 - Bond past month return (bond short-term reversal)
(Pospisil, Zhang (2010))
 - Bond-level measures of risk (predict bond returns positively)
(Bao et al (2015), Israel, Palhares, Richardson (2018), Chordia et al (2017), Chung, Wang, Wu (2019), Bai, Bali, Wen (2021), Dickerson, Mueller, Robotti (2023))
 - Bond ownership concentration (predict bond returns positively)
(Inozemtsev (2023))
- Issuer-level signals (Chordia et al (2017))
 - Firm Asset Growth (predicts bond returns negatively)
 - Market Value of Equity (predicts bond returns negatively)
 - Equity Profitability (predicts bond returns negatively)
 - Equity Momentum (predicts bond returns positively)
 - Equity Past Month Return (predicts bond returns positively)
 - * But many other issuer-level signals do not predict $\mathbb{E}[R]$ (e.g., accruals)
 - * Also, typical signal provides little $\mathbb{E}[R]$ after trading costs

APPENDIX

Existence Conditions for Rational Bubbles (Details) ▸

$$P_t = \sum_{h=1}^{\infty} \mathbb{E}_t \left[\frac{CF_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right] \quad \text{requires} \quad \lim_{h \rightarrow \infty} \mathbb{E}_t \left[\frac{P_{t+h}}{\prod_{\tau=0}^{h-1} \mathbb{E}_{t+\tau}[R]} \right] = 0$$

- Bubbles need to satisfy an important restriction:

- Let $P_t = PV_t + B_t$ with PV_t reflecting the present value of future CF_t

- We have
$$PV_t + B_t = \frac{\mathbb{E}_t [PV_{t+1} + CF_{t+1} + B_{t+1}]}{\mathbb{E}_t [R_{t+1}]}$$

- We also have
$$PV_t = \frac{\mathbb{E}_t [PV_{t+1} + CF_{t+1}]}{\mathbb{E}_t [R_{t+1}]}$$

- Combining these equations, we have
$$B_t = \frac{\mathbb{E}_t [B_{t+1}]}{\mathbb{E}_t [R_{t+1}]}$$

- With $\mathbb{E}_t[R] = \mathbb{E}R$ (for simplicity):
$$B_t = \mathbb{E}_t[B_{t+1}/\mathbb{E}R]$$

- Rational Bubbles:

- $B_t \neq 0$ cannot exist on finite lived assets
- $B_t < 0$ cannot exist if $P_t \geq \bar{P}$ (e.g., limited liability)
- With only $B_t > 0$, Bubbles exist only if always there (Diba, Grossman (1987))
- $B_t \neq 0$ cannot exist in representative agent economy with infinitely lived agent (Santos, Woodford (1997))
- $B_t \neq 0$ can exist in some (but not all) Overlapping Generations (OLG) models (Tirole (1985), Kocherlakota (1992, 2008), Doblas-Madrid (2012), Farhi, Tirole (2012))

Campbell-Shiller Log-Linear PV Identity (Proof)

- From the definition of a gross return with $CF > 0$ (and suppressing the j index):

$$\begin{aligned} \log(R_{t+1}) &= \log(P_{t+1} + CF_{t+1}) - \log(P_t) = \log(P_{t+1}) + \log(1 + CF_{t+1}/P_{t+1}) - \log(P_t) \\ &\downarrow \\ r_{t+1} &= p_{t+1} + \log(1 + e^{cf_{t+1} - p_{t+1}}) - p_t \end{aligned}$$

- From a first-order Taylor approximation:

$$f(cf_{t+1} - p_{t+1}) \approx f(\overline{cf - p}) + f'(\overline{cf - p}) \cdot ((cf_{t+1} - p_{t+1}) - (\overline{cf - p}))$$

- Using $f(cf_{t+1} - p_{t+1}) = \log(1 + e^{cf_{t+1} - p_{t+1}})$ and substituting the result in r_{t+1} :

$$r_{t+1} \approx \kappa_0 + (1 - \kappa) \cdot cf_{t+1} + \kappa \cdot p_{t+1} - p_t$$

$$w/ \quad \kappa = 1/(1 + e^{\overline{cf - p}}) \quad \& \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \cdot \log(1/\kappa - 1)$$

- Isolating p_t and recursively substituting p_{t+h} : (Campbell, Shiller (1989))

$$\begin{aligned} p_t &\approx \kappa_0 + (1 - \kappa) \cdot cf_{t+1} - r_{t+1} + \kappa \cdot p_{t+1} \\ &= (1 + \kappa) \cdot \kappa_0 + (1 - \kappa) \cdot (cf_{t+1} + \kappa \cdot cf_{t+2}) - (r_{t+1} + \kappa \cdot r_{t+2}) + \kappa^2 \cdot p_{t+2} \\ &\vdots \quad \text{keep substituting } p_{t+h} \end{aligned}$$

$$= \frac{\kappa_0}{1 - \kappa} + (1 - \kappa) \cdot \sum_{h=1}^{\infty} \kappa^{h-1} \cdot cf_{t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot r_{t+h} + \lim_{H \rightarrow \infty} \kappa^H \cdot p_{t+H}$$

$$p_t \approx \frac{\kappa_0}{1 - \kappa} + (1 - \kappa) \cdot \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[cf_{t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{t+h}]$$

Campbell-Shiller Log-Linear PV/CF Identity (Proof)

- From the definition of a gross return with $CF > 0$ (and suppressing the j index):

$$\begin{aligned} \log(R_{t+1}) &= \log(P_{t+1} + CF_{t+1}) - \log(P_t) = \log(P_{t+1}) + \log(1 + CF_{t+1}/P_{t+1}) - \log(P_t) \\ &\Downarrow \\ r_{t+1} &= p_{t+1} + \log(1 + e^{cf_{t+1} - p_{t+1}}) - p_t \end{aligned}$$

- From a first-order Taylor approximation:

$$f(cf_{t+1} - p_{t+1}) \approx f(\overline{cf - p}) + f'(\overline{cf - p}) \cdot ((cf_{t+1} - p_{t+1}) - (\overline{cf - p}))$$

- Using $f(cf_{t+1} - p_{t+1}) = \log(1 + e^{cf_{t+1} - p_{t+1}})$ and substituting the result in r_{t+1} :

$$\begin{aligned} r_{t+1} &\approx \kappa_0 + (1 - \kappa) \cdot cf_{t+1} + \kappa \cdot p_{t+1} - p_t \\ &= \kappa_0 + \Delta cf_{t+1} + \kappa \cdot (p_{t+1} - cf_{t+1}) - (p_t - cf_t) \end{aligned}$$

$$w/ \quad \kappa = 1/(1 + e^{\overline{cf - p}}) \quad \& \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \cdot \log(1/\kappa - 1)$$

- Isolating $p_t - cf_t$ and recursively substituting $p_{t+h} - cf_{t+h}$: (Campbell, Shiller (1989))

$$\begin{aligned} p_t - cf_t &\approx \kappa_0 + \Delta cf_{t+1} + r_{t+1} + \kappa \cdot (p_{t+1} - cf_{t+1}) \\ &= (1 + \kappa) \cdot \kappa_0 + (\Delta cf_{t+1} + \kappa \cdot \Delta cf_{t+2}) + (r_{t+1} + \kappa \cdot r_{t+2}) + \kappa^2 \cdot (p_{t+2} - cf_{t+2}) \\ &\vdots \quad \text{keep substituting } p_{t+h} - cf_{t+h} \\ &= \frac{\kappa_0 \cdot (1 - \kappa^H)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \Delta cf_{t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot r_{t+h} + \kappa^H \cdot (p_{t+H} - cf_{t+H}) \end{aligned}$$

$$p_t \approx \frac{\kappa_0 \cdot (1 - \kappa^H)}{1 - \kappa} + cf_t + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta cf_{t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{t+h}] + \underbrace{\kappa^H \cdot \mathbb{E}_t[p_{t+H} - cf_{t+H}]}_{\xrightarrow{H \rightarrow \infty} 0}$$

Economic Significance of Return Predictability (Proof)

- We want to evaluate $\mathbb{E}[\Delta U] = \mathbb{E}[(\omega_t - \omega) \cdot \mu_t] - 0.5 \cdot \mathbb{E}[\gamma \cdot (\omega_t^2 - \omega^2) \cdot \sigma_\epsilon^2]$

with the weights given by $\omega_t = \frac{1}{\gamma} \cdot \frac{\mu_t}{\sigma_\epsilon^2}$ and $\omega = \frac{1}{\gamma} \cdot \frac{\mu}{\sigma_\mu^2 + \sigma_\epsilon^2}$

- First:
$$\begin{aligned}\mathbb{E}[(\omega_t - \omega) \cdot \mu_t] &= \frac{1}{\gamma} \cdot \mathbb{E} \left[\frac{\mu_t^2}{\sigma_\epsilon^2} - \frac{\mu \cdot \mu_t}{\sigma_\mu^2 + \sigma_\epsilon^2} \right] \\ &= \frac{1}{\gamma} \cdot \left(\frac{\mu^2 + \sigma_\mu^2}{\sigma_\epsilon^2} - \frac{\mu^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \right) \\ &= \frac{1}{\gamma} \cdot \left(\frac{\mu^2 + \sigma_\mu^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \cdot \frac{\sigma_\mu^2 + \sigma_\epsilon^2}{\sigma_\epsilon^2} - \frac{\mu^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \right) \\ &= \frac{1}{\gamma} \cdot \left((SR^2 + R^2) \cdot \frac{1}{1 - R^2} - SR^2 \right) \\ &= \frac{1}{\gamma} \cdot \frac{R^2 \cdot (1 + SR^2)}{1 - R^2}\end{aligned}$$

- Second:
$$\begin{aligned}\mathbb{E}[\gamma \cdot (\omega_t^2 - \omega^2) \cdot \sigma_\epsilon^2] &= \frac{1}{\gamma} \cdot \mathbb{E} \left[\frac{\mu_t^2}{\sigma_\epsilon^2} - \left(\frac{\sigma_\epsilon^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \right)^2 \cdot \frac{\mu^2}{\sigma_\epsilon^2} \right] \\ &= \frac{1}{\gamma} \cdot \left(\frac{\mu^2 + \sigma_\mu^2}{\sigma_\epsilon^2} - \left(\frac{\sigma_\epsilon^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \right)^2 \cdot \frac{\mu^2}{\sigma_\epsilon^2} \right) \\ &= \frac{1}{\gamma} \cdot \left(\frac{\mu^2 + \sigma_\mu^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \cdot \frac{\sigma_\mu^2 + \sigma_\epsilon^2}{\sigma_\epsilon^2} - \left(\frac{\sigma_\epsilon^2}{\sigma_\mu^2 + \sigma_\epsilon^2} \right)^2 \cdot \frac{\mu^2}{\sigma_\epsilon^2} \right) \\ &= \frac{1}{\gamma} \cdot \left((SR^2 + R^2) \cdot \frac{1}{1 - R^2} - (1 - R^2) \cdot SR^2 \right) \\ &= \frac{1}{\gamma} \cdot \frac{R^2 + SR^2 \cdot (1 - (1 - R^2)^2)}{1 - R^2}\end{aligned}$$

Treasury Bond Strip Present Value Identity (Proof)

- The log return on an H -year Treasury bond strip is:

$$\begin{aligned} \log(\$R_{b,t+1}^{(H)}) &= \log(\$P_{b,t+1}^{(H-1)} / \$P_{b,t}^{(H)}) \\ &\downarrow \\ r_{b,t+1}^{(H)} &= \$P_{b,t+1}^{(H-1)} - \$P_{b,t}^{(H)} \end{aligned}$$

- Recursively substituting the bond price (and $\$P_{b,t}^{(0)} = 0$): (Campbell, Ammer (1993))

$$\begin{aligned} \$P_{b,t}^{(H)} &= -\$r_{b,t+1}^{(H)} - \$r_{b,t+2}^{(H-1)} + \$P_{b,t+2}^{(H-2)} \\ &= -\$r_{b,t+1}^{(H)} - \$r_{b,t+2}^{(H-1)} - \$r_{b,t+3}^{(H-2)} + \$P_{b,t+3}^{(H-3)} \\ &\vdots \quad \text{keep substituting } \$P_{b,t+h}^{(H-h)} \\ &= -\sum_{h=1}^H \$r_{b,t+h}^{(H-h+1)} \end{aligned}$$

- Taking expectations and using $y_{b,t}^{(H)} = -(1/H) \cdot \$P_{b,t}^{(H)}$

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\$r_{b,t+h}^{(H-h+1)}]$$

- Using $r_{b,t}^{(h)} = \$r_{b,t}^{(h)} - \Delta\pi_t$

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

- Using $r_{b,t}^{(h)} = x_{b,t}^{(h)} + r_{b,t}^{(1)}$

$$y_{b,t}^{(H)} = \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[x_{b,t+h}^{(H-h+1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[r_{b,t+h}^{(1)}] + \frac{1}{H} \cdot \sum_{h=1}^H \mathbb{E}_t[\Delta\pi_{t+h}]$$

Variation in Corporate Bond Prices (Proof)

- Let $C_{j,t}$ reflect the cash flow of bond j so that $R_{j,t+1} = (P_{j,t+1} + C_{j,t+1})/P_{j,t}$
- Assume that:
 - cb has the same structure as tb (maturity and cash flows)
 - $C_{tb,t}/C_{cb,t} = \exp(l_{cb,t})$ (i.e., recovery rate is the same for coupons and principal)
 - If cb defaults then the proceeds are invested in tb
- Then, a log-linear approximation for returns implies (Campbell, Shiller (1989))

$$\log(R_{j,t+1}) \approx \text{const} + \Delta C_{j,t+1} + \kappa \cdot (p_{j,t+1} - C_{j,t+1}) - (p_{j,t} - C_{j,t})$$
- The difference between log returns ($xr_{cb} = \log(R_{cb}) - \log(R_{tb})$) is:

$$\begin{aligned} xr_{cb,t+1} &\approx \text{const} - \kappa \cdot (p_{cb,t,t+1} - p_{cb,t,t+1}) + (p_{cb,t,t} - p_{cb,t,t}) - (\Delta C_{tb,t+1} - \Delta C_{cb,t+1}) \\ &= \text{const} - \kappa \cdot p_{scb,t+1} + p_{stb,t} - (\Delta C_{tb,t+1} - \Delta C_{cb,t+1}) \\ &= \text{const} - \kappa \cdot p_{scb,t+1} + p_{stb,t} - l_{cb,t+1} \end{aligned}$$
 - the second line uses assumptions (1)+(3) $\Rightarrow C_{cb,t} = C_{tb,t}$ outside of default
 - the third line uses assumptions (2)+(3) $\Rightarrow \Delta C_{tb,t} - \Delta C_{cb,t} = l_{cb,t}$
- Isolating $p_{scb,t}$ and recursively substituting $p_{scb,t+h}$, we have

$$\begin{aligned} p_{scb,t} &\approx \text{const} + xr_{cb,t+1} + l_{cb,t+1} + \kappa \cdot p_{scb,t+1} \\ &= \text{const} + (xr_{cb,t+1} + \kappa \cdot xr_{cb,t+2}) + (l_{cb,t+1} + \kappa \cdot l_{cb,t+2}) + \kappa^2 \cdot p_{scb,t+2} \\ &\vdots \\ &\quad \text{keep substituting } p_{scb,t+h} \\ &= \text{const} + \sum_{h=1}^H \kappa^{h-1} \cdot xr_{cb,t+h} + \sum_{h=1}^H \kappa^{h-1} \cdot l_{cb,t+h} \end{aligned}$$

and taking expectations we have (Nozawa (2017))

$$p_{scb,t} \approx \text{const} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[xr_{cb,t+h}] + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[l_{cb,t+h}]$$